SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers trained at one megapixel fail at 4K: full attention meets unfamiliar token counts and position offsets, and its cost grows quadratically. Training-free fixes edit positional encodings or restrict the keys each query sees; AReA, the strongest single-pass baseline, applies one key schedule to every head. Each head's balance between local detail and global layout, we find, is stable and measurable at the training resolution. Scale-Covariant Attention (SCA) emulates a larger target inside the training distribution, fits one local/global mixing weight per head and noise level in closed form against the model's own full attention, and reuses the table at the target. Block-centre global attention and an exact native-size sliding window give every image query the native number of keys at linear cost, and a short global-only opening fixes the layout. For FLUX.1-dev at 4096^2, SCA is the strongest single-pass training-free operator we tested. It beats every shared-sampler baseline on all per-image metrics (ImageReward 1.23 against 0.92 for AReA on 191 prompts) and, of eight methods, has the lowest mean FID and patch FID, the best crop MUSIQ, and the shortest time: 71 s per image, $2.85\times$ faster than AReA. Multi-stage recipes with their own budgets (HRDiT, UltraImage) score higher on learned preference metrics; doubling SCA's steps narrows that gap. Gains hold on a second prompt set, two more seeds, and FLUX.2-dev, and are mixed at 2048×4096.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained at about one megapixel. At 4096×4096 the image-token count grows 16-fold and full-attention cost about 256-fold, and every attention layer sees token counts and position offsets outside its training distribution. Outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025; Rajabi et al., 2026a); in FLUX.1-dev at 4096^2, full attention collapses into stripes or a flat field (Figure 1).

Training-free fixes edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a), the most recent of the second kind, attends to keys subsampled onto a native-size grid for the first 8 of 28 steps, then to windows that shrink to native size, and shares this schedule across heads. Head-specific receptive fields are also established: HRDiT assigns per-head scopes under a cost budget (Xue et al., 2026), and UltraImage concentrates attention per head using entropy (Zhao et al., 2025). We ask a narrower question: can a head's local/global output mixture be fitted at native resolution and transferred to larger images?

We answer it by emulation. To mimic a target s times larger, we shrink both a native-size neighbourhood and a downsampled view of the whole image by s, so the image-to-window ratio matches the target while full attention stays in distribution and serves as ground truth. A convex mix of the two views fits each head well. The fitted weight differs strongly between heads, is stable across tested prompt splits (correlation 0.99), and transfers: the table fitted at emulated scale 2 explains scale 4 with 2.4% more output error than the scale-4 table, and a 1024^2 table explains 1280^2 with 0.8% more (Section 3). We call this measured transfer scale covariance: the local view follows a

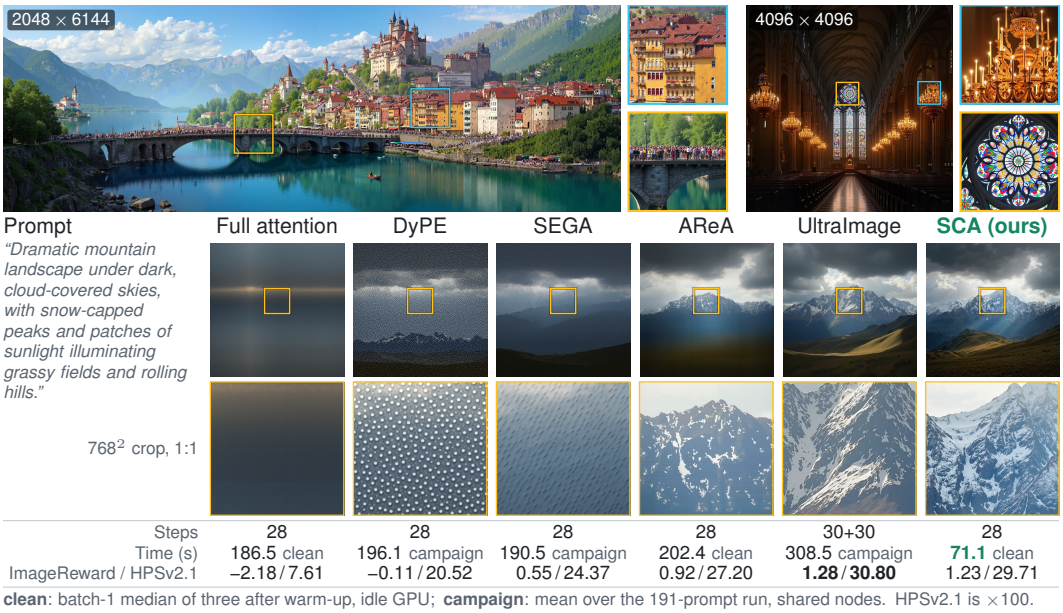

Figure 1: **SCA keeps layout and fine detail at 4096^2 and 2048×6144 without training and is the fastest method shown.** Top: curated SCA outputs; tiles show same-colour boxes at native pixels. Bottom: one curated prompt at 4096^2, seed 0; UltraImage uses its own two-stage recipe. Times are clean batch-1 medians or campaign means (one GH200); metrics are 191-prompt means.

law fixed in tokens, the global view a law fixed in image fractions, and over the tested scales the weight between them stays nearly constant.

Scale-Covariant Attention (SCA) turns this measurement into an attention operator (Figure 2). Each head uses its calibrated weight to mix two branches: a global branch that attends to block-centre keys on the native grid with rescaled coordinates, and a local branch that is an exact native-size sliding window. The calibrated mix alone does not preserve layout: applied from the first step at 4096^2, it draws five Moai statues for a prompt asking for one (Figure 5). SCA therefore runs the global branch alone for the first 8 of 28 steps (the opening phase), as a frequency argument suggests (Argument 1). Our contributions are:

- **Calibration by emulation.** A closed-form procedure that emulates the target scale inside the training distribution and fits each head's receptive-field mix to the model's own full attention from one pass of statistics (Proposition 3). On FLUX.1-dev the mix is head-specific, prompt-stable, and transfers across scale and calibration resolution.
- **The SCA operator.** A per-head, per-noise-level mix of two branches whose image query-key offsets stay in the trained range (Proposition 1) at a cost linear in the token count (Proposition 2). One attention layer at 4096^2 takes 18.7 ms, full attention 87.3 ms.
- **Paired evidence against seven baselines.** Full attention, AReA, DyPE, SEGA, and SigMa under one FLUX.1-dev protocol, and UltraImage and HRDiT under their own multi-stage recipes, with paired bootstrap intervals, ablations, failure galleries, and checks on HRGen-1K, extra seeds, and FLUX.2-dev. Against AReA on 191 prompts at 4096^2, SCA wins 11 of 15 metrics with paired 95% intervals excluding zero, ties the other four (KID, patch FID, patch KID, Inception score), and runs $2.85\times$ faster.

2 Scale-Covariant Attention

2.1 Setting and notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, $[n] = \{0, \ldots, n-1\}$, and applies joint attention in each

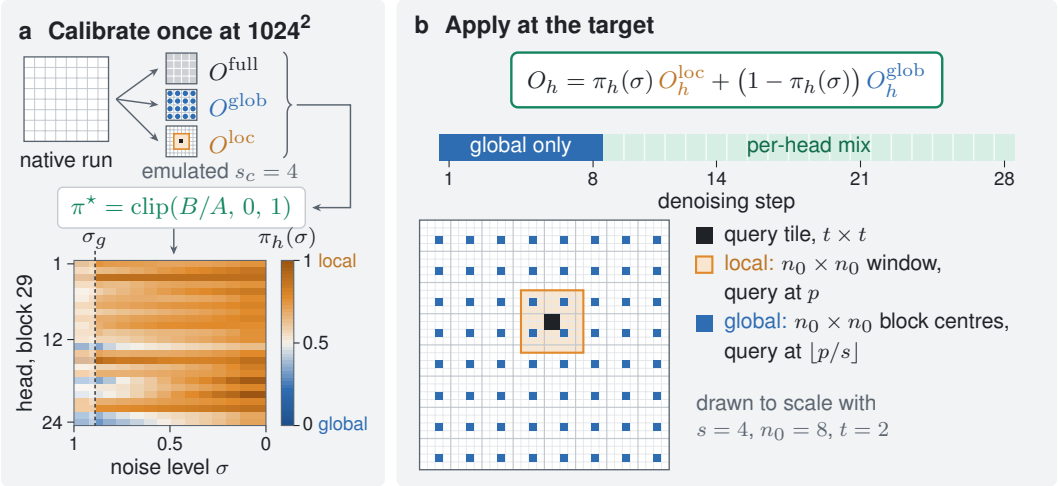

Figure 2: **SCA mixes two native-size branches with per-head weights calibrated once at 1024^2.** (a) $\pi^\star$ minimizes the mix's squared error to full attention. (b) FLUX.1 uses $n_0 = 64$, $t = 8$; steps 1 to 8 form the opening phase.

block (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$ (FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u') \in \mathcal{K}} \mathrm{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v. \tag{1}$$

Full attention is $O_i^{\mathrm{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j \in \mathcal{T} \cup \mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1 - \sigma)x_0 + \sigma\epsilon$. Splitting the keys into a near and a far set gives exactly $O_i^{\mathrm{full}} = m_i O_i^{\mathrm{near}} + (1 - m_i) O_i^{\mathrm{far}}$, where O_i^{near} and O_i^{far} attend only to their own set and m_i is the softmax mass on the near set (Lemma 1): every head is a query-dependent mix of a near and a far field. SCA approximates this mix with two separately normalized operators and one weight per head. Because both branches include text keys and their image key sets can overlap, the identity does not decompose SCA exactly.

2.2 Two operators with native-range image geometry

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\, s_W b_2 + \lfloor s_W/2 \rfloor)$, and the coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$:

$$\mathcal{K}^{\mathrm{glob}} = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b \in \mathcal{G}_0}, \qquad O_i^{\mathrm{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\mathrm{glob}}\big), \tag{2}$$

with $c(p_i)$ replaced by 0 for text queries. This is AReA's first phase with the pooling factor fixed at s. It models content whose extent is a fixed fraction of the image: layout, object count, identity.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. With window side $w = \min(n_0, H)$, the window of tile τ starts at $a(\tau) = \mathrm{clip}\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid. With $\mathcal{W}(i)$ the $w \times w$ window of i's tile,

$$\mathcal{K}_i^{\mathrm{loc}} = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{p_j \in \mathcal{W}(i)}, \qquad O_i^{\mathrm{loc}} = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\mathrm{loc}}\big), \tag{3}$$

an exact sliding window with the original positions. It models content whose extent is fixed in tokens: edges, texture, fine structure.

Proposition 1 (Trained position range)**.** *Assume exact integer position coordinates and $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator and every image query-key offset of the*

local operator is at most $n_0 - 1$ per axis, and each image query sees $T + n_0^2$ keys in either operator, as in full attention on the native grid.

Proposition 2 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N+T)(T+N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T+w^2)$. For fixed n_0 and T the total is $O\big(N(T+w^2+N_0)\big)$, linear in N.*

Proofs are in Appendix A. Proposition 1 assumes exact coordinates; the frozen FLUX.1 implementation rounds positions in bf16 beyond 256 tokens per axis (Appendix C). Native-range geometry also does not make hidden states in-distribution, and text-image offsets in the local operator equal p_i, as in full attention at the target (Remark 1). At 4096^2 SCA scores 7.2 times fewer query-key pairs than full attention, 8.4 times fewer with the opening phase (Table 4).

2.3 Per-head mixing and the opening phase

For an image query of head h in block l at noise level σ,

$$O_i \;=\; O_i^{\mathrm{glob}} \;\text{ if } \sigma > \sigma_g, \qquad O_i \;=\; \pi_{lh}(\sigma)\, O_i^{\mathrm{loc}} + \big(1 - \pi_{lh}(\sigma)\big)\, O_i^{\mathrm{glob}} \;\text{ otherwise,} \tag{4}$$

and text queries use O_i^{glob}. The weight $\pi_{lh}(\sigma)$, written $\pi_h(\sigma)$, stands in for the near-field mass m_i; one weight per head is cheap and a few prompts determine it.

Argument 1 (A frequency-based explanation for the opening phase). At noise level σ a spatial frequency band has signal-to-noise ratio $\big((1-\sigma)/\sigma\big)^2 S(f)$ for image spectrum S. If latent power falls with frequency, only low frequencies carry signal at high noise, and a window of w tokens spans less than one period of any frequency below $1/w$. Independently attended windows may then repeat local structures instead of forming one large object, while a global branch supplies context across windows. Hidden states also carry information from earlier layers, so this is a mechanistic hypothesis, not an impossibility result (Appendix A).

Output matching cannot see this failure: a tokenwise squared error can favour local detail without preserving layout. We have not measured per-band signal-to-noise ratios of FLUX.1 latents, but a 16-token window gives smaller and more numerous copies than a 64-token one, as predicted (Figure 5). We set $\sigma_g = 0.89$ (the first 8 of 28 steps at maximum shift, as in AReA); gating 13 steps gives a 4×4 block mosaic, because all queries of a block share one coarse position.

2.4 Calibration by output matching

The weights must come from where the model is trustworthy, which the target is not. On an in-distribution $n \times n$ grid ($n = 64$ at 1024^2), emulated scale s_c uses subsampling factor s_c, window side n/s_c, and tile side $\mathrm{round}(tn/(s_c n_0))$, so that the ratio of grid to window is s_c, as at a target of scale s_c. The sampler runs unmodified full attention; at every block and step both emulated operators are evaluated on the same queries, keys, and values. For block l, head h, and noise bin β, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \tag{5}$$

and the squared error of the mix with weight π is $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form calibration and transfer cost)**.** *In each cell (l, h, β), J is minimized on $[0,1]$ by $\pi^\star = \mathrm{clip}(B/A, 0, 1)$ ($\pi^\star = 1$ if $A = 0$). Any other weight $\pi' \in [0,1]$ (fitted at another scale or resolution, per layer, shared, or a hard switch) has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$.*

So the per-head table has the lowest output error of any rule constant within a cell, and replacing it costs an A-weighted squared distance. This holds at the emulation scale; Section 3 measures how well tables transfer between scales and resolutions. The same sums score every alternative rule without another forward pass. We use 14 noise bins of width $1/14$ and interpolate linearly between the mean noise levels of occupied bins at inference. One pass collects emulated scales 2 and 4. A target with factors (s_H, s_W) uses the table whose scale is nearest $\max(s_H, s_W)$ in log space, here always 4. The local operator is one FlexAttention call (Dong et al., 2024) over tile-major tokens with the window as an exact block mask (Algorithm 1, Appendix C).

3 WHAT THE HEADS WANT

We calibrate FLUX.1-dev on 32 prompts at 1024^2 and 16 at 1280^2, written for this purpose and disjoint from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\text{full}}\|^2$, pooled over blocks, heads, and steps.

Heads differ. Over the 1368 heads, the standard deviation of $\pi_h(\sigma)$ is 0.24 to 0.26 at every noise level (emulated scale 2). Half of the (head, noise level) cells lie in $[0.2, 0.8]$, 11.7% exceed 0.95, and 2.9% are below 0.05 (Figure 2a; full tables in Figure 6, Appendix B). The mean weight is 0.64 to 0.74, so the typical head leans local but still draws on the global branch.

The table is stable across tested prompt splits. Tables fitted on disjoint halves of the prompts correlate at 0.989 over all cells (0.986 at scale 4; 0.974 for 16 prompts at 1280^2), and each half's table scored on the other half gives a relative error of 0.0576 against 0.0574 in sample. Tables from 4, 8, or 16 prompts also generate similarly (Section 4.3).

The weights transfer across the tested scales and resolutions. The scale-2 table applied at scale 4 gives a relative error of 0.205 against 0.200 for the scale-4 table, still below every shared rule fitted at scale 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. Over all pairs of calibration targets, a transferred table adds at most 5.45% error, while each target's own per-layer fit adds 12 to 24%. The two resolutions emulate scale 2 with windows of 32 and 40 tokens, so this transfer is consistent with weights that depend mainly on the image-to-window ratio.

Per-head beats coarser rules; output error misses layout. Relative to the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4; per-layer tables reach 0.071 and 0.223, one shared weight per noise level 0.077 and 0.230 (Figure 7). In output energy the best hard switch never switches to global (Appendix B), so the opening phase deliberately overrides the fit (Argument 1).

4 EXPERIMENTS

Protocol. All FLUX.1-dev comparisons share AReA's sampler: 28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, bf16, tiled VAE decoding, and seed label 0 for every prompt. The main prompts are the Aesthetic-4K evaluation split with 4K references (Zhang et al., 2025a): 191 prompts after excluding four development prompts, with a fixed 46-prompt subset for ablations. Sizes, written height×width, are 4096^2, 2048×4096, and 2048×6144 (191 prompts each), 6144^2 (100 prompts, [TODO: E2]), and 8192^2 (12 exploratory prompts). Baselines are full attention, AReA from its released code (Appendix F), DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), and UltraImage (Zhao et al., 2025) on all 191 prompts, plus SigMa (Zhao et al., 2026) and HRDiT (Xue et al., 2026) on the 46-prompt subset. UltraImage runs its released guided two-stage recipe (30 steps per stage, guidance 3.5 then 7) and HRDiT its progressive recipe (30, 17, and 10 steps at 1024^2, 2048^2, and 4096^2); both are system-level comparisons with their own budgets. We report preference and alignment scores (ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022), distribution scores (FID, KID, patch FID/KID; Heusel et al., 2017), and no-reference quality (MUSIQ on the full image and on 1024^2 crops, MUSIQ_c; CLIP-IQA; ARNIQA; Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024), with 95% paired bootstrap intervals over prompts. SCA uses $t = 8$, $w = 64$, $\sigma_g = 0.89$, and the scale-4 table from 32 prompts at 1024^2. Each process uses one NVIDIA GH200 120GB. Table 7 and Appendices D and E define every setting and metric.

4.1 MAIN RESULTS

Against AReA. At 4096^2, SCA beats AReA on 11 of 15 metrics with paired 95% intervals excluding zero, including every preference, alignment, and no-reference quality metric (Table 1): ImageReward $+0.31$ $[0.23, 0.39]$, HPSv2.1 $+2.50$, PickScore $+0.73$, CLIP $+1.45$, and MUSIQ_c $+10.9$, all with 95% paired intervals above zero over 2000 prompt draws (Figure 10, Appendix I). FID falls by 7.6 $[2.9, 12.8]$ under a 1000-draw reference-image-cluster bootstrap; corrected patch FID, patch

Table 1: **Among single-pass methods, SCA has the best mean on every metric shown, and it is the fastest.** 4096^2, 191 Aesthetic-4K prompts, FLUX.1-dev. HPSv2.1 $\times 100$; MUSIQ_c: MUSIQ on 1024^2 crops; FID_p: patch FID. **Bold** / underline: best / second best. Seconds per image on one GH200: clean is the batch-1 median after warm-up, campaign the mean over this run on shared nodes; n/m: not measured. Multi-stage recipes lead on preference scores (HRDiT: Table 2); SCA keeps the lowest FID and FID_p and the highest MUSIQ_c (Table 8). [TODO: HRDiT, 191 prompts]

		Time (s)↓		Image	Pick					
Method	Steps	clean	campaign	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID↓	FID_p ↓
Single pass, shared sampler (28 steps, guidance 4.5)										
Full attention	28	186.5	196.5	−2.18	7.61	17.94	15.38	33.2	312.7	252.4
DyPE	28	n/m	196.1	−0.11	20.52	20.49	24.84	50.4	206.2	172.7
SEGA	28	n/m	190.5	0.55	24.37	21.64	26.93	50.4	165.4	126.0
AReA	28	202.4	261.7	0.92	27.20	22.35	27.67	50.4	146.0	103.4
SCA (ours)	28	**71.1**	**71.5**	1.23	29.71	23.09	**29.11**	**61.2**	**138.4**	**99.8**
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	308.5	**1.28**	**30.80**	**23.32**	28.80	41.8	142.7	110.6
HRDiT	30+17+10	?	?	?	?	?	?	?	?	?

Table 2: **On a 46-prompt panel with all eight methods, SCA leads the single-pass methods, and multi-stage recipes lead on preference scores.** Columns as in Table 1; campaign times from this panel. Against SCA at 28 steps (Tables 20 and 21): $^+$ significantly better, $^\circ$ 95% paired interval includes zero, unmarked significantly worse. **Bold**: best. 46-image FID is exploratory; budget runs were scored without whole-image FID (n/a). 42 steps: sized to HRDiT's campaign time; progressive: HRDiT's schedule with SCA. [TODO: HRDiT clean timing; 42-step, progressive]

		Time (s)↓		Image	Pick					
Method	Steps	clean	campaign	Reward	HPSv2.1	Score	CLIP	MUSIQ_c	FID↓	FID_p ↓
Single pass, shared sampler (28 steps, guidance 4.5)										
Full attention	28	186.5	191.6	−2.21	7.00	17.73	14.89	33.6	370.1	292.4
DyPE	28	n/m	204.2	−0.23	20.24	20.25	24.86	50.9	282.4	251.5
SigMa	28	n/m	185.5	−0.77	18.16	19.67	22.95	50.4	289.5	255.4
SEGA	28	n/m	189.7	0.47	24.82	21.56	27.36	51.3	243.4	198.3
AReA	28	202.4	241.0	0.83	27.76	22.31	27.85	52.0	212.7°	168.4°
SCA (ours)	28	**71.1**	**71.6**	1.12	29.98	22.96	**29.41**	62.5	**206.4**	**168.0**
Multi-stage, own recipe and budget										
UltraImage	30+30	n/m	322.0	1.26°	31.54$^+$	23.31$^+$	28.77	42.9	213.8°	182.2
HRDiT	30+17+10	?	106.5	**1.37**$^+$	**31.60**$^+$	**23.42**$^+$	29.38°	54.1	216.9°	178.1
SCA with a larger budget										
56 steps	56	n/m	140.1	1.22$^+$	30.99$^+$	23.12$^+$	29.09°	**64.7**$^+$	n/a	n/a
guidance 7	28	n/m	71.9	1.14°	29.24	22.69	28.98°	63.4°	n/a	n/a
42 steps	42	?	?	?	?	?	?	?	?	?
progressive	30+17+10	?	?	?	?	?	?	?	?	?

KID, whole-image KID, and Inception score have intervals spanning zero (Appendix E). SCA takes 71.1 s per image, a 2.6× speedup over full attention and 2.85× over AReA. Full attention collapses into stripes and is significantly worse than AReA on every per-image metric. Crops in Figures 1 and 3 show the baselines' smoothing, speckle, and tiling.

Other baselines. Under the shared protocol, SCA also improves every per-image metric over DyPE and SEGA on 191 prompts (Figure 10) and over all five shared-protocol baselines, SigMa included, on the 46-prompt panel (Tables 2 and 20). This fixes the FLUX.1-dev backbone and sampler rather than each baseline's tuned defaults (Appendix G). Multi-stage recipes lead on learned preference scores. On 191 prompts UltraImage is higher on HPSv2.1 (by 1.10) and PickScore (by 0.24) with an unresolved ImageReward difference, while SCA is higher on CLIP and MUSIQ_c (by 19.4) with lower FID and patch FID (Tables 16 and 19). On the panel HRDiT reaches ImageReward

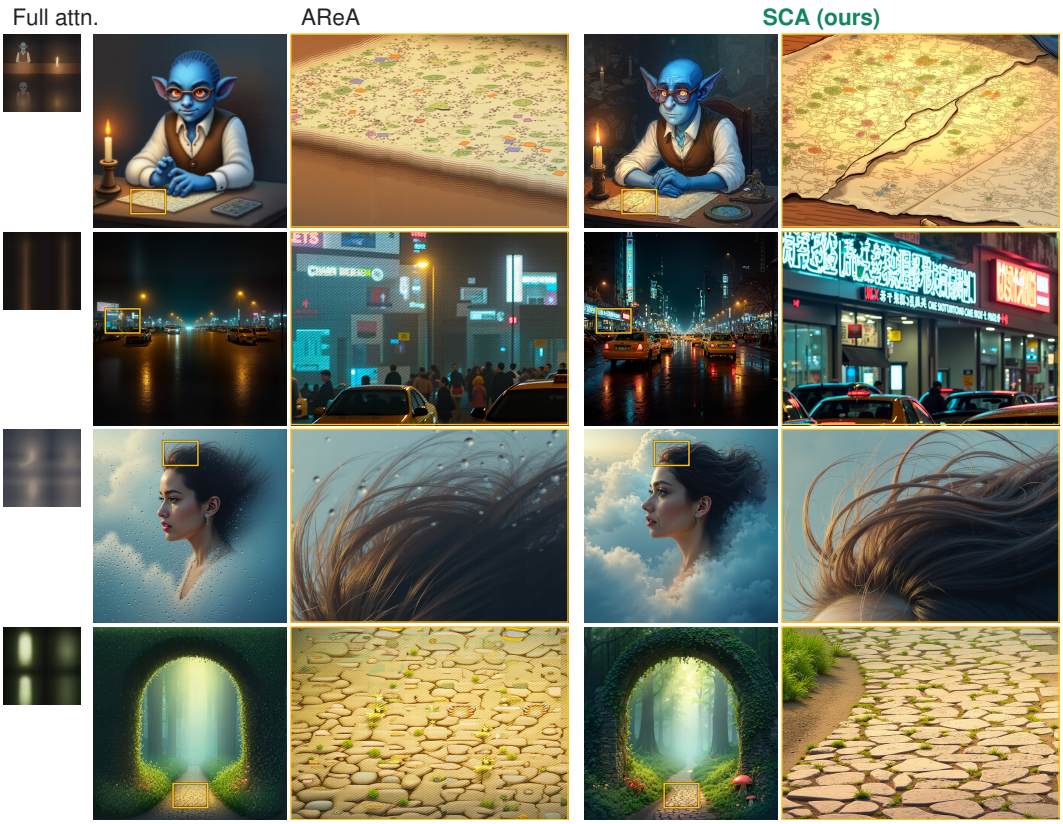

Figure 3: **At 4096^2 full attention collapses, AReA blurs or tiles fine structure, and SCA keeps both layout and detail.** Seed 0; four curated prompts, hand-picked for variety from 191 evaluation prompts. Beside each AReA and SCA image is the same native 736×512 crop, auto-placed at maximal edge energy averaged over the three methods. Uncurated samples: Figure 12.

1.37, HPSv2.1 31.60, and PickScore 23.42 against 1.12, 29.98, and 22.96 for SCA, which keeps higher MUSIQ_c and lower patch FID. With 56 steps (140.1 s per image) SCA reaches 1.22, 30.99, and 23.12, narrowing the gap without closing it (Appendix J). Campaign means are 71.6 s for SCA, 106.5 s $[103.2, 111.0]$ for HRDiT ($1.5\times$), and 308.5 to 322.0 s for UltraImage; clean batch-1 timing exists only for full attention, AReA, and SCA.

Other sizes. Gains depend on the aspect ratio (Table 10). At 2048×6144 (factors 2 and 6), SCA beats AReA significantly on ImageReward ($+0.27$ $[0.15, 0.40]$), FID (-19.8 $[-24.4, -11.8]$), and eight other metrics, and loses on aesthetic score (-0.15). At 2048×4096 (factors 2 and 4) results are mixed: ImageReward (-0.00 $[-0.08, 0.07]$) and FID are unresolved, HPSv2 and both MUSIQ scores favour SCA, and aesthetic score, CLIP-IQA, and patch FID/KID favour AReA, with or without the scale-2 table (Appendix L). On 12 exploratory prompts at 8192^2, paired scores favour SCA on preference and alignment, MUSIQ differences are unresolved, and grid and stripe artifacts remain (Appendix K). [TODO: E2: 6144^2 metrics.]

Latency. SCA's attention cost is linear in N (Proposition 2); at 4096^2 one attention layer takes 18.7 ms against 87.3 ms for full attention. End to end, SCA is the fastest of the methods measured at each size (Figure 4, Table 5), with 161.6 and 287.6 s at 6144^2 and 8192^2 against 482.3 and 952.0 s for AReA ($2.99\times$ and $3.31\times$). From 2048^2 to 8192^2 the pixel count grows $16\times$ and SCA's time $15.8\times$. [TODO: E3: finish the full-attention 6144^2 timing group.]

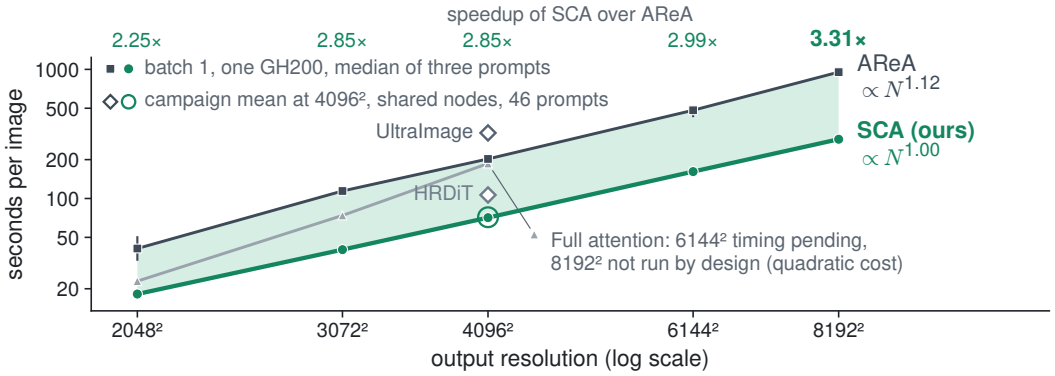

Figure 4: **SCA is the fastest of the methods measured at each size, and its lead over AReA grows to 3.31× at 8192².** Filled markers: batch-1 seconds per image on one GH200 (median of three prompts, whiskers min to max); full attention scales as $N^{1.51}$ (N: image tokens). Hollow markers: 46-prompt campaign means at 4096^2 on shared nodes; HRDiT runs 30, 17, 10 steps at 1024^2, 2048^2, 4096^2, UltraImage two 30-step stages. HRDiT's clean batch-1 timing is pending.

Table 3: **Every ablation significantly lowers HPSv2.1 and PickScore.** 4096^2, 46 prompts. First row: SCA's absolute values; other rows: paired difference after changing one component, in **bold blue** / orange when significantly better / worse (95% interval excludes zero). HPSv2.1 and CLIP-IQA are $\times 100$. Mix err.: relative output error at emulated scale 4. Full results: Table 11.

Variant	Mix err.	ImageReward	HPSv2.1	PickScore	CLIP-IQA	MUSIQ$_c$
SCA (reference)	0.200	1.12	29.98	22.96	71.3	62.5
Weight rule after the opening phase						
local only ($\pi = 1$)	0.261	−0.36	−2.86	−1.20	−4.6	−0.3
one shared $\pi(\sigma)$	0.230	−0.09	−1.83	−0.38	−7.7	−10.4
per-layer $\pi_\ell(\sigma)$	0.223	−0.05	−0.70	−0.16	−0.9	−2.4
Global branch and opening phase						
global only ($\pi = 0$)	0.627	−0.40	−5.66	−1.44	−27.0	−11.7
mean pooling	n/a	−1.52	−9.75	−2.60	−10.1	−4.2
no opening phase (v0)	0.200	−1.33	−4.07	−2.18	**+2.9**	+2.5

4.2 Ablations

Table 3 changes one component at a time (examples in Figure 17). **The opening phase is essential.** Without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$ and subjects repeat; CLIP-IQA and ARNIQA rise because each copy is sharp. **The per-head mix carries global context after the opening phase.** Local-only attention in the last 20 steps lowers ImageReward by 0.36 $[0.21, 0.52]$, HPSv2.1 by 2.86, PickScore by 1.20, and CLIP score by 1.93. **Per-head weights beat shared and per-layer weights, in the order output error predicts.** One shared weight per noise level (mix error 0.230) lowers ImageReward by 0.09 $[0.01, 0.17]$ and MUSIQ$_c$ by 10.4; per-layer weights (0.223) still lose significantly on HPSv2.1, PickScore, ARNIQA, and MUSIQ$_c$. **Neither branch works alone, and block centres beat block means.** Global only loses on every metric, and mean pooling blurs the global keys (ImageReward -1.52). Band-guard ablations, including hard per-head routing, are in Appendix M.

4.3 Generality and sensitivity

Table 15 (Appendix I) summarizes these checks and points to their intervals. On the first 200 HRGen-1K prompts (Rajabi et al., 2026a), SCA raises ImageReward over AReA from 0.138 to 0.780 and improves all ten per-image metrics (the set has no reference images). On the 46-prompt subset, seeds 1 and 2 keep ImageReward gains of 0.305 and 0.355. Calibrating on 4, 8, or 16 instead of 32 prompts changes ImageReward by at most 0.012, and no per-image metric differs sig-

nificantly for the four-prompt table, so modest calibration data suffices on this prompt distribution. Gate thresholds 0.85 and 0.93 each lose on some metrics, so the three thresholds do not identify a metric-independent optimum. On FLUX.2-dev, a table calibrated on 16 native-resolution prompts lifts ImageReward over full attention at 4096^2 from 0.748 to 1.204 and MUSIQ_c from 38.5 to 60.0 on 50 prompts; nine of ten per-image intervals favour SCA, and mean denoising-plus-decoding time falls from 444.0 to 220.2 s (full attention is the only baseline here; Appendix G). [TODO: E9: FLUX.1-Krea-dev.]

5 Related work

Training-free high-resolution methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024; Qiu et al., 2024), add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025), or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). Closest to SCA, AReA shares one schedule of pooled global attention and shrinking windows across heads (Rajabi et al., 2026a); ResDiT spectrally fuses position-scaled global and native-size patch attention between global-only and local-only phases (Ma et al., 2025); FreeScale fuses attention outputs by frequency (Qiu et al., 2024). HRDiT picks per-head scopes by loss sensitivity under a cost budget (Xue et al., 2026), and UltraImage concentrates attention per head by entropy, noting stable local/global head roles before us (Zhao et al., 2025). Our local branch is neighbourhood attention (Hassani et al., 2022), ToDo attends to downsampled keys without rescaling coordinates (Smith et al., 2024), and sparse attention in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022) varies windows, spans, or key scales across layers or heads. Head specialization, two branches, and a global opening phase are thus established; SCA adds a calibration procedure: emulate the target scale at native resolution, fit a per-head, per-noise-level output mixture in closed form, and transfer it to an operator with native key counts.

6 Limitations and conclusion

Limitations. SCA does not win everywhere. At 2048×4096 it has no resolved ImageReward advantage over AReA and loses on several other metrics. Multi-stage recipes keep higher preference scores, HRDiT at 1.5 times SCA's campaign time and UltraImage at over four times. Small extra figures can still appear late in sampling, the exploratory 8192^2 gallery retains grid artifacts, and one development prompt keeps doubled limbs in every variant (Appendix Q). The opening phase was chosen from development generations, not by the output-error objective. Transfer across emulated scales supports, but cannot prove, the hypothesis that the weights depend only on the image-to-window ratio. The full 191-prompt comparison uses FLUX.1-dev and one seed per prompt; extra seeds, SigMa, and HRDiT cover 46 prompts, and the FLUX.2-dev comparison covers 50 prompts against full attention only. SigMa's dev-matched profile gives lattice-degraded images, which may reflect our port or profile rather than the method. We did not run ResDiT, for which we could not verify official code, and our guided UltraImage configuration does not cover its direct-extrapolation variant, so superiority over those settings is unestablished. Low-sample FID estimates and automatic preference models cannot replace human evaluation, and the intervals are not adjusted for multiple comparisons. [TODO: E2, E3, E9: 6144^2 comparison, large-size latency, Krea.]

Conclusion. How much each head relies on a neighbourhood fixed in tokens versus a view of the whole image can be measured once at the training resolution and reused at 4K. With that table and a global-only opening, SCA is the strongest and fastest single-pass training-free operator we tested for FLUX.1-dev at 4096^2, at about a third of AReA's time; multi-stage recipes lead on learned preference metrics.

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A PROOFS

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}}O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}}O^{\text{near}} + Z^{\text{far}}O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The key count is $T + w^2 = T + n_0^2$ when $n_0 \leq \min(H, W)$. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys and w^2 window keys. The sum is at most $(2N + T)(T + w^2 + N_0)$, which is $O(N(T + w^2 + N_0))$ for fixed n_0, T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs: a query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\text{loc}} = O^{\text{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\text{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \le 0$, so $A\pi^\star - B \ge 0$ and $\pi' - \pi^\star \ge 0$. If $\pi^\star = 1$ then $B \ge A$, so $A\pi^\star - B \le 0$ and $\pi' - \pi^\star \le 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell, such as a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere, is a feasible π' in every cell, so its total error is at least the per-head table's, by $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$ or more. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

The frequency argument in more detail. Argument 1 rests on two facts and one assumption. The first fact is that the noisy latent is a sum of signal and independent white noise, so a band's signal-to-noise ratio scales as $((1-\sigma)/\sigma)^2 S(f)$; at $\sigma = 0.89$ the factor is 0.015, and at $\sigma = 0.5$ it is 1. The second is that a window of w tokens cannot distinguish a component of period longer than w from a smooth trend within the window, so windows cannot coordinate a layout at that scale except through what earlier layers mixed into their keys. The assumption is that natural-image latent spectra fall with frequency, which we have not measured for FLUX.1 latents. Together they predict that windowed attention fails first and worst at high noise and for objects larger than the window, which matches the duplication we observe (Figure 5) and the effect of a 16-token window.

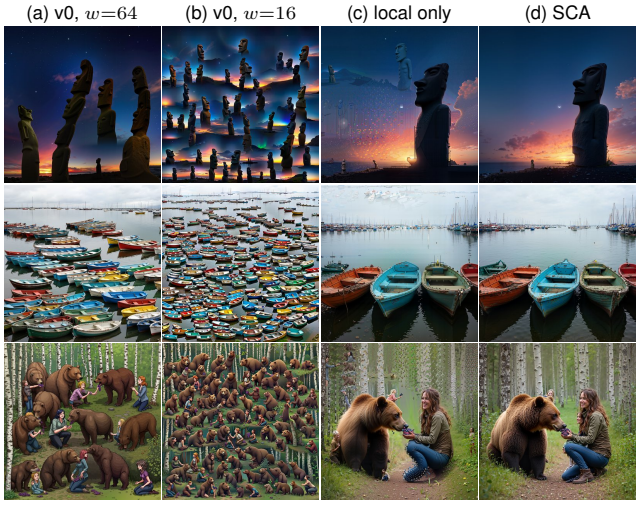

Figure 5: **Windows at high noise duplicate subjects.** 4096^2, three development prompts. (a) v0, the calibrated mix from the first step: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) With the opening phase but local-only attention afterwards ($\pi = 1$), ghost copies and grid speckle remain. (d) SCA.

B Calibration details and statistics

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\text{full}} - O^{\text{loc}}\|^2$, $\sum \|O^{\text{full}}\|^2$, and the query count, in float64, summed over image queries and channels (`sca/calib.py`). The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure 8); within a block, the

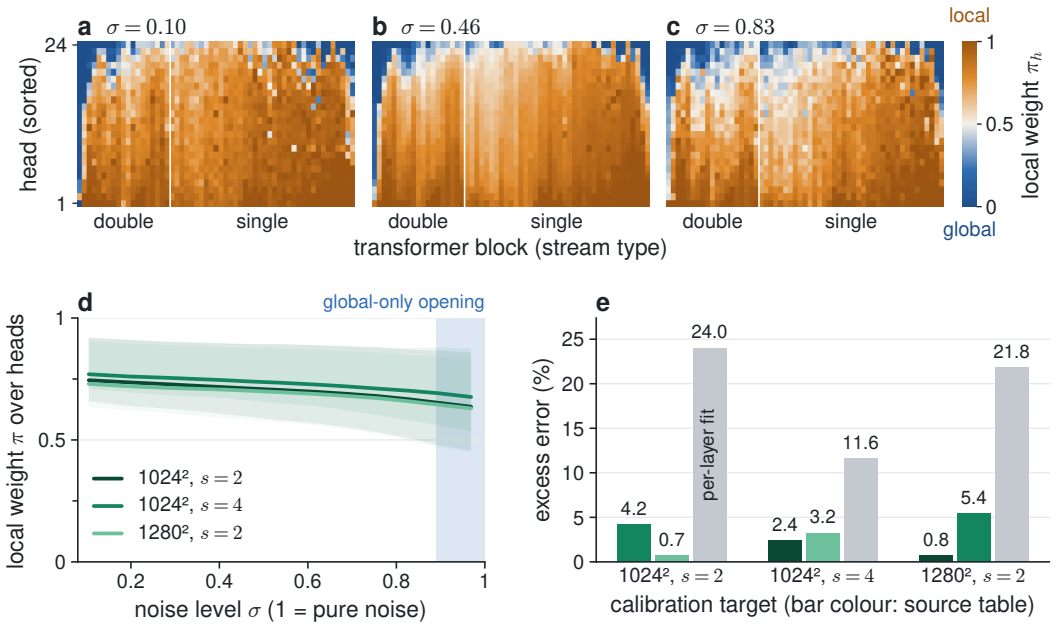

Figure 6: **Heads differ widely, yet a calibrated table transfers across scale and resolution.** (a to c) $\pi_h(\sigma)$ of the scale-4 table used at inference, heads sorted per block. (d) Mean and interquartile range over the 1368 heads for three calibrations (32 prompts at 1024^2, 16 at 1280^2). (e) Excess error over the target's per-head table: at most 5.45% when transferred, 12 to 24% for its per-layer fit.

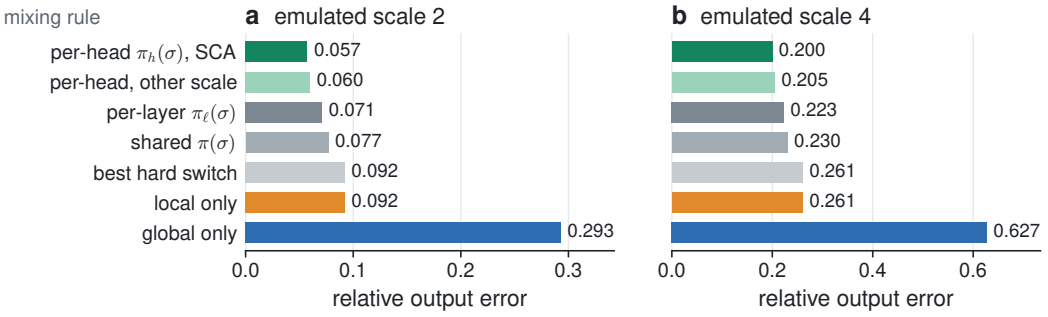

Figure 7: **Per-head weights explain full attention best, and transfer across scales.** Relative output error of each mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts. "Other scale" applies the scale-4 table at scale 2 and vice versa; the best hard switch has $\sigma^\star = 1$ and so equals local only. In-sample and held-out errors agree within 0.0003.

per-head weights span a median range of 0.66 to 0.84 at every noise level (Figure 9). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 4.2 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution.

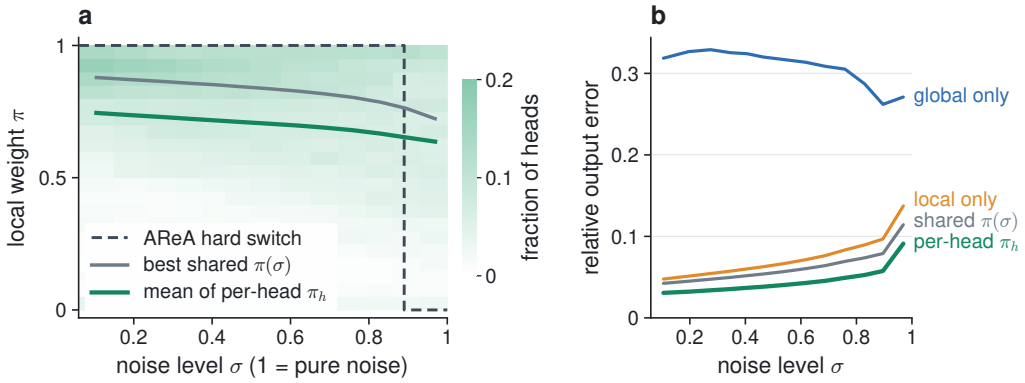

Figure 8: **In output energy, no noise level prefers the global branch.** (a) Fraction of the 1368 heads per π bin and noise bin (shading), with the mean of $\pi_h(\sigma)$, the best shared $\pi(\sigma)$, and AReA's schedule read as a weight; emulated scale 2, 32 prompts. (b) Relative output error per noise bin: local alone beats global alone everywhere, and per-head mixing beats both.

Calibration budget. We merge the stored sufficient statistics from the first 4, 8, or 16 calibration prompts and rerun the 46 evaluation prompts with each resulting table. Table 18 reports generation-level differences from the 32-prompt table. The test reuses calibration observations but uses separate prompts for generation evaluation; it is not a held-out output-error benchmark.

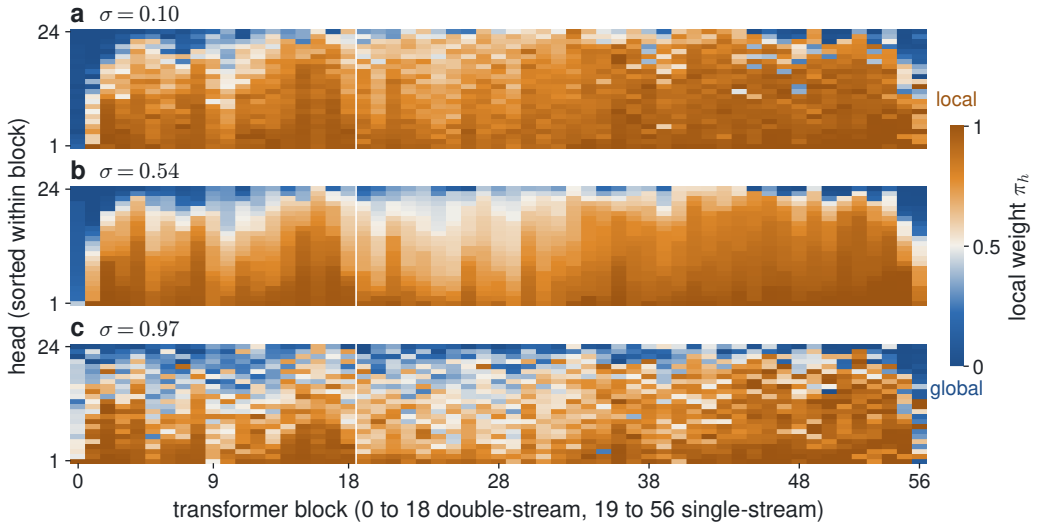

Figure 9: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the scale-2 table for all 57 blocks and 24 heads of FLUX.1-dev, 32 prompts; heads are sorted within each block by their weight at $\sigma = 0.54$, in the same order in every panel. The median within-block range is 0.66 to 0.84. Figure 6 shows the scale-4 table used at inference.

C Implementation details and cost

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge; text keys come first. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. On GH200, 8×16 query blocks with 8×8 key blocks were fastest

Algorithm 1 SCA: calibration (once per model) and one attention layer at sampling time.

Calibrate(prompts, s_c) on an in-distribution $n \times n$ grid:
sample with full attention; at every block l, step σ, and head h:
  $O^{\text{glob}}, O^{\text{loc}} \leftarrow$ Eqs. 2, 3 with subsampling s_c, window n/s_c, tile $tn/(s_c n_0)$
  add $\|O^{\text{loc}} - O^{\text{glob}}\|^2$ and $\langle O^{\text{full}} - O^{\text{glob}}, O^{\text{loc}} - O^{\text{glob}} \rangle$ to $A_{lh\beta(\sigma)}$, $B_{lh\beta(\sigma)}$ ▷ float64
return $\pi_{lh\beta} \leftarrow \text{clip}(B_{lh\beta}/A_{lh\beta}, 0, 1)$ ▷ Proposition 3
Layer(Q, K, V, σ) at target factors (s_H, s_W):
$O^{\text{glob}} \leftarrow$ Eq. 2 for all queries ▷ block-centre keys, coarse positions
if $\sigma > \sigma_g$ **then return** O^{glob} ▷ opening phase
$O^{\text{loc}} \leftarrow$ Eq. 3 for image queries ▷ one FlexAttention call
return $[\, O^{\text{glob}}_{\mathcal{T}},\ \pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}_{\mathcal{G}} \,]$ ▷ interpolated table; fp32 mix

at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Numerical checks. The 18 CPU tests in `tests/test_sca.py` use float32 to check full-mode agreement with stock FLUX attention, both branches against brute-force Equations 2 and 3, full-grid and factor-1 identities, calibration when the local branch is exact, the gate, and FlexAttention against an explicit gather reference. A separate GPU check in `scripts/bench_attn.py` compares the gather and FlexAttention implementations in bf16: the recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our frozen runs retain these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry, but does not restore exact RoPE coordinates: the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384 and its scale-6 coarse coordinate becomes 64; at a 512-token axis, index 511 rounds to 512 and its scale-8 coarse coordinate also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 1 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0–255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would require a separate controlled experiment; the reported runs are unchanged.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t; neither case arises at the sizes we evaluate.

Table 4: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the local branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Size ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
128×384	2048×6144	49.6	14.0	3.5

End-to-end latency. Table 5 comes from one matched sweep on one GH200 (`results/latency_curve_summary.csv`). At each size, each method generates one excluded warm-

Table 5: **Seconds per image**, batch 1, idle GH200, 28 steps with text encoding and VAE decoding, median of three after warm-up. All populated cells come from one matched sweep that is still running; full attention at 8192^2 is outside it. **?**: pending.

Size (N tokens)	Full attention	AReA	SCA
2048^2 (16,384)	22.9	41.0	**18.2**
3072^2 (36,864)	73.9	114.3	**40.2**
4096^2 (65,536)	186.5	202.4	**71.1**
6144^2 (147,456)	**?**	482.3	**161.6**
8192^2 (262,144)	not run	952.0	**287.6**

up image and then three timed images in the same process, at batch 1 and 28 steps; we report the median. At 4096^2 the medians are 186.5 s for full attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 71.1 s for SCA (71.0, 71.1, 71.1). At 2048^2 and 3072^2 they are 22.9 and 73.9 s for full attention, 41.0 and 114.3 s for AReA, and 18.2 and 40.2 s for SCA. At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s and SCA 161.6 and 287.6 s, so SCA reduces measured time by 66.5% and 69.8%. An earlier 4096^2 benchmark, kept as a separate repeat (`results/latency_4096_batch1.csv`), gave 184.6 s for full attention (184.1, 184.7, 184.6), 212.6 s for AReA (207.3, 225.8, 212.6), 88.1 s for SCA + band guard (88.1, 98.7, 88.1; single-pass compiled band), and 71.0 s for SCA final (71.0, 83.1, 70.9). On the shared campaign nodes, per-image times at 4096^2 were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (48 s) than in the clean measurement (17 s). At the non-square sizes we have only campaign timings, with both methods on the same pool of shared nodes (mean over 191 prompts): SCA takes 38.1 s and AReA 112.4 s at 2048×4096, and 55.8 s and 150.5 s at 2048×6144. [TODO: E3: finish the full-attention 6144^2 timing group, Table 5.]

D HYPERPARAMETERS

Table 6 lists every setting of the main runs.

Table 6: **All hyperparameters** of the main runs (code commit `2d11740`), produced by `scripts/sample.py --method sca --pi-table calib/k1_1024.npz --gate-sigma 0.89`. Appendix G gives the settings that differ for UltraImage, HRDiT, SigMa, and FLUX.2-dev.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed 0 per prompt, shared across methods; tiled VAE decoding
Local branch	tile t, window w	8, 64 (clamped at borders), original positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin), $57 \times 24 \times 14$, scale-4 entry
	interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only)
Calibration	prompts, resolution	32 at 1024^2 (16 more at 1280^2 for transfer)
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	hardware	one GH200; complete calibration runtime was not retained
Band guard	(not in final method)	average-pool $p = 16$, bilinear upsampling

E EVALUATION PROTOCOL AND METRICS

Table 7: **Evaluation protocol.** The FLUX.1-dev attention comparisons share sampler settings and prompt seeds. UltraImage and HRDiT use their own multi-stage protocols; FLUX.2-dev uses its own native-resolution schedule (Appendix G). Prompt counts exclude four development prompts. Table 6 lists the hyperparameters; the paragraphs below define the metrics.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) [TODO: E9]; FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; UltraImage, HRDiT, and FLUX.2-dev exceptions are detailed in Appendix G.
Sizes	4096^2, 2048×4096, 2048×6144 (191 prompts each); 6144^2 (100 prompts, [TODO: E2]); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 191 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), 200 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (two-stage defaults; Zhao et al., 2025); on the 46-prompt subset also SigMa (Zhao et al., 2026) and HRDiT (released progressive protocol; Xue et al., 2026).
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120 GB model. Clean latency uses batch 1; campaign timings are marked separately.

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and window and are excluded from generation evaluation, leaving 191. Distribution metrics retain all 195 official reference rows. The 46-prompt ablation subset was fixed before any ablation was run. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use source rows 0–199, with ids `hrgen_0000`–`hrgen_0199`. The 48 calibration prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100 \max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics and reference correction. FID (Heusel et al., 2017) and KID compare the generated images (191, or 46 on the subset) with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The reference rows contain 168 distinct image contents; repeated rows retain their official weights. Patch metrics use four 1024^2 crops per row (764 generated and 780 reference crops). An audit found that the original feature-bank assembly keyed reference crops by resolved image path, so repeated images reused the last row's crop features. We corrected the lookup to retain each row's intended cached crops. Whole-image FID/KID are unchanged; all patch point estimates reported here use the corrected assembly. Original CSVs remain archived for provenance. Inception score uses full images at 299^2, with 10 splits.

Intervals. Per-image intervals are 95% paired percentile bootstraps over prompt ids: original summaries use 1000 draws; separately recomputed comparisons use 2000 draws with seed 0. For the main SCA–AReA FID/KID and patch-FID/KID comparison, 1000 draws resample generated prompt ids jointly across methods and resample the 168 distinct reference-image identities, keeping all rows and crops from each sampled identity together. This retains the 195-row weighted point estimate while accounting for repeated reference images. The resulting FID difference is -7.63 $[-12.79, -2.94]$; patch FID is -3.60 $[-7.39, 1.15]$ and patch KID ($\times 10^3$) is -2.27 $[-4.73, 0.12]$. An earlier version of this analysis reported a significant patch-KID improvement; under the corrected evaluation its interval includes zero. The whole-image and patch FID/KID comparisons at both non-square sizes and on the 46-prompt panel, the added baselines' patch comparisons, and the 56-step patch-FID comparison also use reference-image clusters. Other whole-image FID/KID intervals are labelled historical reference-row bootstraps; they have not been recomputed with image-identity clusters. Historical patch intervals are not attached to corrected estimates and are omitted where recomputation is incomplete. FID's finite-sample bias can place point estimates outside percentile intervals. Intervals are unadjusted for multiple comparisons and measure prompt/reference sampling uncertainty, not variability over independent model training runs.

F Reproducing AReA

We run AReA's released code with its FLUX.1 defaults (block-centre pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our shared protocol gives 0.92, so cross-paper numbers are not comparable, while our direct SCA–AReA comparison holds these settings fixed. UltraImage and FLUX.2-dev use separate protocols (Appendix G). AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; the matched sweep measures 202.4 s and 186.5 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge; neither appears in AReA's paper, which does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. [TODO: E9: AReA with its default FLUX.1-Krea-dev checkpoint, against recalibrated SCA.]

G Other baselines and FLUX.2-dev

DyPE and SEGA. We use the existing baseline harness frozen at commit `c05ebdc`, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-derived seeds as SCA. These settings isolate the attention/position treatment on the checkpoint used in our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; the separate Krea comparison remains pending. Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation through `scripts/run_ultraimage.py`. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes preparation when not cached and the two-stage call; main-stage and preparation times are retained separately in sidecars. Each call reseeds both the generator and the global RNG, including the VAE sampling path, from the prompt id and seed label. Thus seed labels are paired, but the two-stage method does not share an identical latent trajectory with the one-stage methods. The wrapper preserves an upstream scheduler behaviour: assigning `scheduler.config.shift` does not update

the scheduler's internal shift in the installed diffusers version, so both stages retain shift 3.0. This is a reproduction caveat, not a newly tuned setting. We report this system-level comparison separately from the common-sampler attention comparisons.

HRDiT. We run the released progressive protocol of the official repository (commit `f3be935`) on FLUX.1-dev in fp16: 30 steps at 1024^2 with guidance 3.5, then 17 steps at 2048^2 with guidance 4.5 and 10 steps at 4096^2 with guidance 6, 57 steps in all, with latent upsampling and the shipped per-head scope plan and attention settings. Its sampling budget therefore differs from the shared protocol, and we report it with the multi-stage recipes. HRDiT's global Torch random state is reset per prompt because its Gaussian blur draws a random width.

SigMa. The dev-matched profile runs the released code (commit `fbe3a86`) on FLUX.1-dev under the shared sampler: 28 steps, guidance 4.5, bf16, YaRN positions, and the shift capped at its maximum from the target sequence length. SigMa's paper states FLUX.1-dev with guidance 3.5, while its released command line hardcodes FLUX.1-Krea-dev with guidance 4.5; we ran only the profile that matches our protocol. The outputs are finite, fully decoded 4096^2 images, but the images we inspected show a regular lattice of dots, as DyPE's do at 4096^2 on FLUX.1-dev. This may reflect our port or profile rather than the method, so the SigMa row should not be read as the method's best result.

Adapter for HRDiT and SigMa. Both run on the 46-prompt subset through one adapter with campaign per-prompt CPU generators, resident models, and tiled VAE decoding. These differ from the upstream command-line seed, offload, and decode choices, so the runs are controlled adaptations, not reproductions of the authors' tables. Campaign time includes text encoding, decoding, and finite-value checks after every step, which add synchronization overhead; it excludes model loading and is not a clean latency benchmark.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and calibration rule with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions preserve the time and text axes while coarsening the two spatial axes. Calibration uses 16 prompts at 1024^2, 14 noise bins, and emulated scales 2 and 4; sampling uses the scale-4 table. Both full attention and SCA use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55; we therefore do not interpret this comparison as a default-pipeline benchmark. The gate is 0.959 so that the first 8 steps use only global attention. The 50 evaluation prompts are the first 50 Aesthetic-4K prompts remaining after development-prompt exclusion. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding, and exclude text encoding and model loading. At emulated scale 4, fitted per-head relative output error is 0.1790, versus 0.1889 for per-layer weights and 0.1924 for local-only attention; split-half correlation is 0.950 and cross-half error 0.1794. These are calibration diagnostics, not generation ablations of the individual components on FLUX.2.

H Extended results with intervals

Tables 9 and 10 give every metric with its paired interval for the main comparisons, Tables 11 to 13 the ablation intervals, and Table 14 the non-square calibration check of Appendix L; Appendix I pairs the added baselines against SCA. All are populated from archived evaluation CSV files. Per-image means were checked against the corresponding per-image records; distribution-level estimates use archived cached features and corrected reference-row crop assembly (Appendix E). Set-level metrics (FID, KID, their patch versions, and Inception score) are computed on the whole set. The final SCA FID/KID and patch-FID/KID intervals resample reference-image clusters; other FID/KID intervals retain the historical reference-row bootstrap unless a corrected image-cluster interval is available, and Inception-score intervals resample only generated images.

Table 8: **At 4096^2, SCA beats AReA on every per-image metric at about a third of its latency.** Aesthetic-4K with FLUX.1-dev, 191 prompts (100 at 6144^2). HPS, CLIP-IQA, and ARNIQA are $\times 100$; higher is better except latency and FID. Bold: best per size. * / † on full-attention and SCA rows: significantly better / worse than AReA (95% paired interval excludes zero). Latency at 4096^2: seconds per image, batch 1, idle GH200, median of three. ‡Mean campaign time on shared nodes, not clean latency. $^\circ$UltraImage uses its two-stage defaults. **?**: pending.

Method	Lat. (s)↓	Image Reward	Aesth.	CLIP	Pick Score	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096 × 4096, scale factors (4, 4)													
Full attention	186.5	$-2.18^\dagger$	$4.51^\dagger$	$15.38^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.61^\dagger$	$312.7^\dagger$	252.4	$22.9^\dagger$	$33.2^\dagger$	$49.0^\dagger$	$54.5^\dagger$
DyPE	$196.1^\ddagger$	−0.11	5.73	24.84	20.49	25.52	20.52	206.2	172.7	36.3	50.4	59.5	60.1
SEGA	$190.5^\ddagger$	0.55	6.03	26.93	21.64	27.15	24.37	165.4	126.0	37.4	50.4	66.8	64.8
UltraImage$^\circ$	$308.5^\ddagger$	**1.28**	**6.50**	28.80	**23.32**	**29.97**	**30.80**	142.7	110.6	33.5	41.8	72.7	**72.3**
AReA (released code)	202.4	0.92	6.28	27.67	22.35	28.64	27.20	146.0	103.4	39.4	50.4	67.5	67.3
SCA + band guard	88.1	1.20^*	6.46^*	28.87^*	22.90^*	29.40^*	28.95^*	142.3	**98.8**	41.4^*	61.0^*	70.9^*	70.8^*
SCA (final)	**71.1**	1.23^*	6.45^*	**29.11***	23.09^*	29.77^*	29.71^*	**138.4***	99.8	**45.3***	**61.2***	**72.9***	71.2^*
2048 × 4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.51**	**22.67**	28.38	**27.66**	144.8	**112.8**	51.8	61.6	**74.8**	**73.0**
SCA (final)	**38.1**†	**1.13**	$6.21^\dagger$	28.21	22.64	**28.55***	27.46	**143.8**	$118.3^\dagger$	**55.9***	**65.9***	$73.5^\dagger$	72.8
2048 × 6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.15**	26.49	21.51	26.41	23.35	163.6	**106.5**	42.9	53.1	67.3	69.3
SCA (final)	**55.8**‡	**0.66***	$6.00^\dagger$	**27.01**	**21.74***	**26.54***	**23.48**	**143.8***	110.1	**48.4***	**60.6***	**69.0***	**70.2***
6144 × 6144, scale factors (6, 6), 100 prompts [TODO: E2]													
AReA (released code)	?	?	?	?	?	?	?	?	?	?	?	?	?
SCA (final)	?	?	?	?	?	?	?	?	?	?	?	?	?

Table 9: **All metrics at 4096^2**, 191 prompts, paired against AReA (released code). Δ is method minus AReA with its 95% paired bootstrap interval (1000 draws); **blue** / *orange*: significantly better / worse than AReA. Set-level intervals follow the policy stated after Table 8; corrected patch values without recomputed intervals have no significance marks. ‡Mean campaign wall-clock on shared nodes; Table 5 gives clean latency. Sources: `results/final4k_nb_*`, `final4k_*`, `final4k_full_*`.

Metric	AReA	SCA final	Δ [95% CI]	SCA + band	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.917	1.229	**+0.312** [+0.240, +0.384]	1.198	**+0.281** [+0.208, +0.358]	−2.178	−3.095 [−3.216, −2.978]
Aesthetic	6.28	6.45	**+0.17** [+0.12, +0.22]	6.46	**+0.18** [+0.12, +0.24]	4.51	−1.77 [−1.87, −1.67]
CLIP score	27.67	29.11	**+1.45** [+1.06, +1.82]	28.87	**+1.20** [+0.79, +1.61]	15.38	−12.28 [−13.10, −11.44]
PickScore	22.35	23.09	**+0.73** [+0.62, +0.86]	22.90	**+0.54** [+0.40, +0.71]	17.94	−4.41 [−4.61, −4.18]
HPSv2 (×100)	28.64	29.77	**+1.12** [+0.95, +1.31]	29.40	**+0.75** [+0.53, +0.99]	20.75	−7.89 [−8.23, −7.54]
HPSv2.1 (×100)	27.20	29.71	**+2.50** [+2.17, +2.88]	28.95	**+1.75** [+1.31, +2.22]	7.61	−19.59 [−20.39, −18.71]
MUSIQ	39.4	45.3	**+5.8** [+4.1, +7.5]	41.4	**+2.0** [+0.0, +3.9]	22.9	−16.5 [−18.0, −15.0]
MUSIQ_c	50.4	61.2	**+10.9** [+9.5, +12.4]	61.0	**+10.6** [+9.2, +12.3]	33.2	−17.1 [−18.8, −15.4]
CLIP-IQA (×100)	67.5	72.9	**+5.4** [+4.0, +6.8]	70.9	**+3.3** [+1.8, +4.9]	49.0	−18.6 [−20.4, −16.7]
ARNIQA (×100)	67.3	71.2	**+3.9** [+3.0, +4.9]	70.8	**+3.5** [+2.4, +4.6]	54.5	−12.8 [−14.3, −11.2]
Inception score	6.42	6.52	+0.09 [−0.47, +0.49]	6.10	−0.33 [−0.88, +0.18]	3.83	−2.60 [−3.14, −1.83]
FID↓	146.0	138.4	**−7.6** [−12.8, −2.9]	142.3	−3.7 [−9.6, +1.0]	312.7	+166.7 [+145.2, +170.9]
KID↓ ($\times 10^3$)	7.11	6.09	−1.02 [−3.12, +1.16]	5.48	−1.63 [−3.84, +0.51]	163.97	+156.86 [+139.17, +174.94]
FID_p ↓	103.4	99.8	−3.6 [−7.4, +1.2]	98.8	−4.7	252.4	+149.0
KID_p ↓ ($\times 10^3$)	11.44	9.17	−2.27 [−4.73, +0.12]	9.67	−1.77	171.16	+159.72
Seconds per image‡ ↓	261.7	71.5	**−190.2** [−203.0, −176.8]	120.0	**−141.7** [−155.8, −127.1]	196.5	**−65.2** [−79.2, −50.5]

Table 10: **All metrics at the non-square sizes**, 191 prompts each, SCA final paired against AReA (released code). Conventions as in Table 9. Corrected patch intervals use reference-image clusters where available; absent intervals are omitted. Sources: `results/final_2048x4096_*`, `final_2048x6144_*`.

	2048 × 4096, factors (2, 4)			2048 × 6144, factors (2, 6)		
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.133	1.130	−0.003 [−0.079, +0.073]	0.389	0.664	**+0.275** [+0.150, +0.400]
Aesthetic	6.41	6.21	−0.21 [−0.26, −0.15]	6.15	6.00	−0.15 [−0.25, −0.04]
CLIP score	28.51	28.21	−0.30 [−0.70, +0.11]	26.49	27.01	+0.52 [−0.01, +1.07]
PickScore	22.67	22.64	−0.02 [−0.13, +0.09]	21.51	21.74	**+0.23** [+0.05, +0.43]
HPSv2 (×100)	28.38	28.55	**+0.17** [+0.06, +0.28]	26.41	26.54	**+0.13** [+0.00, +0.27]
HPSv2.1 (×100)	27.66	27.46	−0.20 [−0.49, +0.05]	23.35	23.48	+0.13 [−0.27, +0.54]
MUSIQ	51.8	55.9	**+4.1** [+2.7, +5.7]	42.9	48.4	**+5.5** [+3.6, +7.3]
MUSIQ$_c$	61.6	65.9	**+4.3** [+3.3, +5.5]	53.1	60.6	**+7.5** [+6.2, +9.0]
CLIP-IQA (×100)	74.8	73.5	−1.3 [−2.3, −0.4]	67.3	69.0	**+1.6** [+0.2, +3.1]
ARNIQA (×100)	73.0	72.8	−0.2 [−0.9, +0.5]	69.3	70.2	**+0.9** [+0.1, +1.7]
Inception score	6.21	6.65	+0.44 [−0.16, +0.95]	5.36	6.35	**+0.99** [+0.13, +1.22]
FID↓	144.8	143.8	−1.1 [−6.4, +4.3]	163.6	143.8	**−19.8** [−24.4, −11.8]
KID↓ (×10^3)	7.95	7.87	−0.09 [−2.90, +2.97]	18.08	8.46	**−9.62** [−13.94, −5.60]
FID$_p$ ↓	112.8	118.3	+5.5 [+1.9, +9.5]	106.5	110.1	+3.6 [−0.2, +11.3]
KID$_p$ ↓ (×10^3)	14.53	16.48	+1.95 [+0.39, +3.70]	13.66	15.36	+1.70 [−2.22, +6.07]
Seconds per image‡ ↓	112.4	38.1	**−74.3** [−76.6, −72.0]	150.5	55.8	**−94.7** [−96.9, −92.4]

Table 11: **Ablation intervals for Table 3**, 46 prompts at 4096^2, paired against SCA final (2000 bootstrap draws); **blue** / *orange*: significantly better / worse than SCA final. Panels group the variants as in Table 3; the SCA final column gives its absolute values, and the last row of each panel gives patch FID per variant (set-level, no paired interval). Sources: `results/ablnb_*`.

Weight rule after the opening phase				
Metric	SCA final	$\pi = 1$	shared $\pi(\sigma)$	per-layer
ImageReward	1.124	−0.357 [−0.524, −0.207]	−0.090 [−0.173, −0.012]	−0.054 [−0.159, +0.030]
Aesthetic	6.45	−0.21 [−0.35, −0.07]	−0.09 [−0.18, −0.02]	+0.01 [−0.06, +0.07]
CLIP score	29.41	−1.93 [−2.68, −1.20]	−0.38 [−0.78, +0.08]	−0.13 [−0.54, +0.28]
PickScore	22.96	−1.20 [−1.44, −0.99]	−0.38 [−0.51, −0.26]	−0.16 [−0.28, −0.05]
HPSv2 (×100)	29.71	−1.78 [−2.07, −1.50]	−0.96 [−1.19, −0.73]	−0.50 [−0.63, −0.37]
HPSv2.1 (×100)	29.98	−2.86 [−3.35, −2.36]	−1.83 [−2.24, −1.43]	−0.70 [−0.98, −0.41]
MUSIQ	46.1	−3.5 [−6.5, −0.7]	−4.9 [−7.9, −1.9]	+0.7 [−1.6, +3.0]
MUSIQ_c	62.5	−0.3 [−2.5, +1.7]	−10.4 [−12.8, −8.3]	−2.4 [−3.6, −1.1]
CLIP-IQA (×100)	71.3	−4.6 [−6.3, −3.0]	−7.7 [−9.4, −6.0]	−0.9 [−2.0, +0.1]
ARNIQA (×100)	70.5	−1.0 [−2.0, +0.0]	−6.5 [−8.5, −4.8]	−1.5 [−2.3, −0.9]
$\text{FID}_p \downarrow$	168.0	192.2	166.7	167.7

Global branch and opening phase				
Metric	SCA final	$\pi = 0$	mean pool	v0, no gate
ImageReward	1.124	−0.395 [−0.507, −0.295]	−1.518 [−1.847, −1.193]	−1.328 [−1.595, −1.064]
Aesthetic	6.45	−0.76 [−0.87, −0.64]	−0.61 [−0.77, −0.44]	−0.08 [−0.17, +0.01]
CLIP score	29.41	−3.48 [−4.43, −2.59]	−3.59 [−4.76, −2.49]	−1.65 [−2.58, −0.84]
PickScore	22.96	−1.44 [−1.66, −1.22]	−2.60 [−3.00, −2.17]	−2.18 [−2.51, −1.85]
HPSv2 (×100)	29.71	−3.32 [−3.72, −2.94]	−4.04 [−4.66, −3.43]	−1.89 [−2.34, −1.45]
HPSv2.1 (×100)	29.98	−5.66 [−6.35, −4.98]	−9.75 [−11.30, −8.17]	−4.07 [−5.11, −3.04]
MUSIQ	46.1	−3.5 [−6.7, −0.2]	−1.8 [−5.3, +1.9]	−5.6 [−9.0, −2.1]
MUSIQ_c	62.5	−11.7 [−13.6, −9.8]	−4.2 [−8.4, −0.5]	+2.5 [−0.4, +5.5]
CLIP-IQA (×100)	71.3	−27.0 [−30.2, −23.8]	−10.1 [−13.0, −7.1]	**+2.9** [+1.0, +4.9]
ARNIQA (×100)	70.5	−45.1 [−47.6, −42.5]	−4.8 [−6.7, −2.6]	**+4.3** [+2.7, +6.1]
$\text{FID}_p \downarrow$	168.0	237.0	184.6	202.6

Table 12: **Intervals for the band-guard ablations** (Table 23 and further variants), 46 prompts at 4096^2, paired against SCA + band guard, whose absolute values form the first column; **blue** / *orange*: significantly better / worse than this reference. Corrected patch FID is shown without intervals. Sources: `results/abl_*`, `route_*`.

Metric	Reference	no band	no gate	shared $\pi(\sigma)$	per-layer	$\pi = 1$
ImageReward	1.165	−0.041 [−0.109, +0.021]	−0.357 [−0.511, −0.222]	−0.032 [−0.084, +0.015]	−0.024 [−0.055, +0.004]	−0.015 [−0.075, +0.040]
Aesthetic	6.43	+0.01 [−0.05, +0.08]	−0.23 [−0.35, −0.11]	−0.06 [−0.13, +0.00]	+0.03 [−0.01, +0.06]	+0.03 [−0.02, +0.10]
CLIP score	29.01	+0.40 [−0.04, +0.92]	−1.15 [−1.82, −0.47]	−0.09 [−0.53, +0.31]	−0.03 [−0.28, +0.23]	+0.16 [−0.19, +0.53]
PickScore	22.71	**+0.25** [+0.09, +0.41]	−0.86 [−1.13, −0.59]	+0.02 [−0.10, +0.14]	+0.01 [−0.06, +0.08]	**+0.12** [+0.03, +0.22]
HPSv2 (×100)	29.18	**+0.54** [+0.31, +0.75]	−0.49 [−0.87, −0.14]	−0.20 [−0.42, +0.01]	−0.08 [−0.19, +0.04]	**+0.16** [+0.04, +0.29]
HPSv2.1 (×100)	28.81	**+1.17** [+0.71, +1.64]	−1.39 [−2.18, −0.63]	−0.33 [−0.68, +0.01]	−0.01 [−0.19, +0.19]	**+0.55** [+0.30, +0.79]
MUSIQ	43.1	**+2.9** [+0.7, +5.7]	−0.7 [−3.6, +2.4]	+0.5 [−2.5, +4.0]	−0.3 [−1.9, +1.8]	+1.0 [−0.3, +2.6]
MUSIQ_c	62.1	+0.4 [−0.9, +1.6]	−1.0 [−3.4, +1.7]	−5.7 [−7.2, −4.5]	−1.5 [−2.4, −0.7]	**+0.9** [+0.3, +1.4]
CLIP-IQA (×100)	69.6	**+1.7** [+0.3, +3.2]	−0.6 [−2.7, +1.5]	−6.5 [−8.4, −4.7]	−1.2 [−2.1, −0.5]	+0.5 [−0.9, +1.9]
ARNIQA (×100)	69.6	+0.9 [−0.6, +3.1]	+1.6 [−0.4, +4.5]	−4.6 [−7.1, −2.9]	−1.2 [−2.0, −0.6]	+0.6 [−0.1, +1.4]
$\text{FID}_p \downarrow$	166.7	+1.3	+8.9	−4.6	−1.3	+1.6

Table 13: **Band-guard ablations, continued.** Conventions as in Table 12.

Metric	Reference	route 0.5	route 0.3	mean pool	$\pi = 0$	v0
ImageReward	1.165	−0.051 [−0.095, −0.009]	−0.026 [−0.082, +0.024]	−0.848 [−1.083, −0.615]	−0.436 [−0.537, −0.336]	−1.369 [−1.633, −1.113]
Aesthetic	6.43	−0.02 [−0.08, +0.04]	+0.02 [−0.03, +0.07]	−0.39 [−0.54, −0.25]	−0.75 [−0.86, −0.63]	−0.07 [−0.15, +0.03]
CLIP score	29.01	−0.00 [−0.31, +0.28]	+0.13 [−0.23, +0.51]	−2.96 [−4.18, −1.67]	−3.08 [−3.98, −2.16]	−1.25 [−2.02, −0.40]
PickScore	22.71	+0.04 [−0.03, +0.12]	**+0.09** [+0.01, +0.17]	−1.80 [−2.10, −1.49]	−1.19 [−1.41, −0.96]	−1.94 [−2.23, −1.65]
HPSv2 (×100)	29.18	+0.02 [−0.11, +0.15]	+0.11 [−0.01, +0.24]	−2.35 [−2.81, −1.94]	−2.79 [−3.18, −2.45]	−1.35 [−1.74, −0.95]
HPSv2.1 (×100)	28.81	+0.18 [−0.04, +0.40]	**+0.34** [+0.12, +0.58]	−5.81 [−6.96, −4.74]	−4.48 [−5.17, −3.80]	−2.90 [−3.82, −2.01]
MUSIQ	43.1	+1.2 [−0.2, +3.0]	+0.8 [−0.4, +2.5]	−1.8 [−5.5, +1.7]	−0.6 [−4.4, +3.4]	−2.7 [−6.1, +0.7]
MUSIQ_c	62.1	+0.5 [−0.0, +1.1]	**+0.8** [+0.2, +1.4]	−0.8 [−5.0, +3.6]	−11.3 [−13.3, −9.6]	+2.9 [−0.3, +6.2]
CLIP-IQA (×100)	69.6	+0.3 [−1.3, +1.5]	+0.6 [−0.7, +1.7]	−5.4 [−9.3, −1.6]	−25.2 [−29.1, −21.4]	**+4.6** [+2.1, +7.3]
ARNIQA (×100)	69.6	+0.1 [−0.6, +0.8]	+0.6 [−0.1, +1.3]	−1.1 [−3.6, +1.9]	−44.3 [−47.4, −40.7]	**+5.1** [+2.8, +8.1]
$\text{FID}_p \downarrow$	166.7	+1.5	+1.1	+18.7	+70.3	+35.9

Table 14: **Calibration table at 2048×4096**, 46 prompts, paired against SCA final with the scale-4 table; **blue** / *orange*: significantly better / worse than this reference. Sources: `results/ps2_*`.

Metric	SCA, scale-2 table	AReA (released code)
ImageReward	+0.012 [−0.025, +0.050]	+0.027 [−0.141, +0.192]
Aesthetic	−0.00 [−0.03, +0.02]	**+0.29** [+0.16, +0.43]
CLIP score	−0.02 [−0.22, +0.17]	+0.19 [−0.53, +0.91]
PickScore	**+0.05** [+0.00, +0.09]	+0.14 [−0.09, +0.39]
HPSv2 (×100)	**+0.09** [+0.04, +0.13]	−0.19 [−0.44, +0.06]
HPSv2.1 (×100)	**+0.17** [+0.08, +0.26]	+0.39 [−0.21, +1.01]
MUSIQ	+0.0 [−0.5, +0.5]	−3.1 [−7.5, +1.2]
MUSIQ_c	−0.0 [−0.4, +0.2]	−4.1 [−6.7, −1.3]
CLIP-IQA (×100)	−0.6 [−1.2, +0.0]	+1.3 [−0.8, +3.4]
ARNIQA (×100)	−0.2 [−0.6, +0.1]	+0.4 [−1.5, +2.8]

I ADDITIONAL PAIRED INTERVALS

Table 15: **Gains over AReA hold on a second prompt set and two more seeds, and calibration size barely matters.** Paired differences at 4096^2 from the stated reference; HPSv2.1 is $\times 100$ and n counts evaluation prompts. Calibration and gate rows compare with default SCA (32 calibration prompts, $\sigma_g = 0.89$). Tables 17 and 18 give the intervals. **?**: pending.

Setting	n	Reference	ΔImageReward	ΔHPSv2.1	ΔMUSIQ$_c$
HRGen-1K	200	AReA	+0.642	+4.48	+12.71
Seed 1	46	AReA	+0.305	+2.19	+11.58
Seed 2	46	AReA	+0.355	+2.80	+11.04
4 calibration prompts	46	SCA	+0.012	−0.02	+0.09
8 calibration prompts	46	SCA	+0.010	+0.01	−0.03
16 calibration prompts	46	SCA	−0.007	−0.01	−0.11
Gate $\sigma_g = 0.85$	46	SCA	−0.015	−0.85	−2.04
Gate $\sigma_g = 0.93$	46	SCA	−0.004	−0.15	+0.65
FLUX.2-dev	50	Full attention	+0.456	+1.84	+21.44
FLUX.1-Krea-dev	?	AReA	?	?	?

Per-metric intervals are in Tables 16 and 19 for the added baselines, Table 17 for the prompt-set, seed, and model checks, and Table 18 for the calibration-budget and gate checks. Figure 10 plots four of these metrics for the five baselines run on all 191 prompts, signed as SCA minus baseline. Every difference in the tables is method minus the named reference. Positive values favour the method for the ten per-image metrics, and **blue** / *orange* mark intervals that exclude zero in favour of / against the method. These comparisons resample matched prompt ids, with 2000 draws and seed 0; they do not treat multiple crops as independent images. The HRGen-1K sample has 200 prompts, the two extra seeds and sensitivity experiments each have 46, FLUX.2-dev has 50, and the main baseline comparison has 191. Tables use the separately recomputed per-image CSVs; these intervals differ slightly from the original 1000-draw evaluator summaries. No multiplicity correction is applied.

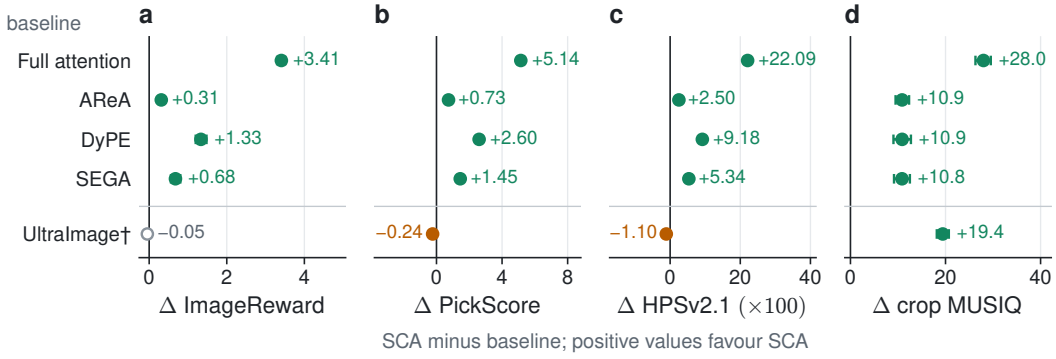

Figure 10: **SCA beats every shared-protocol baseline on all four metrics; against UltraImage the result is mixed.** Mean paired difference on the same 191 prompts at 4096^2; whiskers are 95% prompt-bootstrap intervals (2000 draws, one generation seed, no multiplicity correction); open markers mark intervals that contain zero. †Own two-stage recipe, not an equal-cost comparison.

Pending comparisons. [TODO: E2: 6144^2 intervals.] [TODO: E9: Krea per-metric intervals.]

46-prompt panel. Table 20 pairs each baseline of Table 2 with SCA on the same 46 prompts and seeds. Per-image intervals use 2000 prompt draws with seed 0; FID, KID, and their patch versions use 1000 draws that resample generated prompts and reference-image clusters. With 46 generated images, the whole-image distribution metrics are exploratory. Table 21 gives the matching intervals for the 56-step and guidance-7 SCA runs.

Table 16: **DyPE and SEGA trail SCA on every per-image metric; UltraImage is mixed.** Baseline minus SCA at 4096^2, 191 prompts; **blue** / *orange*: baseline significantly better / worse than SCA. UltraImage uses its own two-stage defaults.

Metric	DyPE	SEGA	UltraImage
ImageReward	−1.334 [−1.478, −1.199]	−0.677 [−0.803, −0.562]	+0.047 [−0.016, +0.111]
Aesthetic	−0.72 [−0.81, −0.63]	−0.42 [−0.49, −0.35]	**+0.06** [+0.01, +0.10]
CLIP score	−4.27 [−4.89, −3.66]	−2.19 [−2.65, −1.77]	−0.31 [−0.63, −0.01]
PickScore	−2.60 [−2.79, −2.41]	−1.45 [−1.60, −1.31]	**+0.24** [+0.14, +0.33]
HPSv2 (×100)	−4.25 [−4.55, −3.97]	−2.61 [−2.85, −2.37]	**+0.21** [+0.05, +0.36]
HPSv2.1 (×100)	−9.18 [−9.95, −8.50]	−5.34 [−5.88, −4.81]	**+1.10** [+0.80, +1.40]
MUSIQ	−9.0 [−11.1, −6.8]	−7.9 [−9.8, −6.0]	−11.7 [−13.6, −9.8]
MUSIQ_c	−10.9 [−12.7, −9.0]	−10.8 [−12.5, −9.2]	−19.4 [−20.7, −18.2]
CLIP-IQA (×100)	−13.4 [−15.3, −11.6]	−6.2 [−7.8, −4.6]	−0.2 [−1.3, +0.8]
ARNIQA (×100)	−11.1 [−12.5, −9.7]	−6.5 [−7.5, −5.4]	**+1.1** [+0.4, +1.8]

Table 17: **The gains hold on HRGen-1K, two more seeds, and FLUX.2-dev.** SCA minus AReA for HRGen-1K and seeds 1 and 2; SCA minus full attention for FLUX.2-dev. All at 4096^2; **blue**: significantly better than the reference.

Metric	HRGen-1K	Seed 1	Seed 2	FLUX.2-dev
ImageReward	**+0.642** [+0.539, +0.755]	**+0.305** [+0.172, +0.458]	**+0.355** [+0.244, +0.470]	**+0.456** [+0.320, +0.598]
Aesthetic	**+0.26** [+0.20, +0.33]	+0.11 [−0.00, +0.23]	**+0.16** [+0.06, +0.25]	**+0.30** [+0.18, +0.42]
CLIP score	**+2.50** [+1.99, +3.03]	**+1.51** [+0.94, +2.12]	**+1.68** [+1.05, +2.33]	+0.38 [−0.38, +1.12]
PickScore	**+1.18** [+1.06, +1.31]	**+0.71** [+0.51, +0.92]	**+0.86** [+0.62, +1.11]	**+0.56** [+0.29, +0.86]
HPSv2 (×100)	**+1.87** [+1.67, +2.07]	**+1.02** [+0.71, +1.31]	**+1.35** [+0.99, +1.71]	**+0.85** [+0.50, +1.21]
HPSv2.1 (×100)	**+4.48** [+3.96, +4.98]	**+2.19** [+1.63, +2.74]	**+2.80** [+2.08, +3.47]	**+1.84** [+1.10, +2.55]
MUSIQ	**+6.7** [+4.7, +8.6]	**+5.0** [+1.0, +9.1]	+3.5 [−0.9, +8.0]	**+16.0** [+11.9, +19.9]
MUSIQ_c	**+12.7** [+11.2, +14.3]	**+11.6** [+9.2, +13.9]	**+11.0** [+8.2, +14.3]	**+21.4** [+18.2, +24.8]
CLIP-IQA (×100)	**+8.1** [+6.5, +9.8]	**+2.2** [+0.0, +4.5]	**+4.0** [+1.7, +6.3]	**+6.6** [+4.7, +8.6]
ARNIQA (×100)	**+7.0** [+5.9, +8.0]	**+3.2** [+1.9, +4.4]	**+3.8** [+2.3, +5.4]	**+7.1** [+5.4, +8.8]

Table 18: **The calibration budget barely matters; a gate at $\sigma_g = 0.85$ lowers seven metrics.** Variant minus default SCA, 46 prompts; **blue** / *orange*: variant significantly better / worse than default SCA. Budgets use the first 4, 8, or 16 of the 32 calibration prompts; the default gate is $\sigma_g = 0.89$.

Metric	4 prompts	8 prompts	16 prompts	$\sigma_g = 0.85$	$\sigma_g = 0.93$
ImageReward	+0.012 [−0.007, +0.029]	+0.010 [−0.004, +0.024]	−0.007 [−0.020, +0.005]	−0.015 [−0.098, +0.064]	−0.004 [−0.097, +0.069]
Aesthetic	+0.01 [−0.00, +0.02]	+0.01 [−0.00, +0.02]	+0.00 [−0.01, +0.01]	−0.19 [−0.26, −0.12]	+0.00 [−0.05, +0.06]
CLIP score	+0.08 [−0.03, +0.19]	+0.05 [−0.04, +0.15]	+0.04 [−0.04, +0.13]	−0.27 [−0.82, +0.26]	−0.64 [−1.03, −0.29]
PickScore	−0.01 [−0.03, +0.01]	+0.00 [−0.01, +0.02]	+0.01 [−0.01, +0.03]	−0.12 [−0.27, +0.02]	−0.22 [−0.38, −0.07]
HPSv2 (×100)	+0.02 [−0.02, +0.05]	+0.02 [−0.00, +0.05]	+0.00 [−0.02, +0.03]	−0.48 [−0.69, −0.28]	+0.02 [−0.16, +0.20]
HPSv2.1 (×100)	−0.02 [−0.07, +0.04]	+0.01 [−0.03, +0.05]	−0.01 [−0.05, +0.04]	−0.85 [−1.22, −0.48]	−0.15 [−0.56, +0.24]
MUSIQ	−0.1 [−0.4, +0.3]	−0.1 [−0.5, +0.2]	**+0.1** [+0.0, +0.3]	−1.9 [−3.5, −0.2]	+0.4 [−1.0, +2.0]
MUSIQ_c	+0.1 [−0.1, +0.3]	−0.0 [−0.2, +0.1]	−0.1 [−0.2, +0.0]	−2.0 [−3.0, −1.1]	+0.6 [−0.4, +1.9]
CLIP-IQA (×100)	+0.2 [−0.1, +0.4]	+0.2 [−0.1, +0.4]	−0.1 [−0.4, +0.1]	−3.2 [−4.8, −1.7]	+0.0 [−1.2, +1.1]
ARNIQA (×100)	+0.1 [−0.1, +0.3]	+0.0 [−0.1, +0.2]	−0.0 [−0.1, +0.1]	−4.0 [−5.9, −2.3]	**+1.8** [+0.7, +3.1]

Table 19: **All three added baselines have significantly higher FID, patch FID, and patch KID than SCA.** Method minus SCA on 191 prompts (negative favours the method); *orange*: significantly worse than SCA. Whole-image FID/KID keep the historical 1000-draw reference-row bootstrap; patch estimates use corrected crops and 1000-draw reference-image cluster intervals.

Metric	DyPE	SEGA	UltraImage
FID↓	+67.8 [+56.9, +79.3]	+27.0 [+21.0, +38.4]	+4.4 [+1.2, +10.5]
KID↓ ($\times 10^3$)	+41.49 [+32.04, +52.45]	+4.98 [+0.95, +10.11]	+0.31 [−1.69, +2.12]
FID_p ↓	+72.9 [+59.6, +81.6]	+26.2 [+17.9, +34.0]	+10.7 [+6.5, +14.7]
KID_p ↓ ($\times 10^3$)	+49.57 [+40.36, +59.66]	+13.76 [+8.95, +19.22]	+7.48 [+5.43, +9.83]

Table 20: **On the 46-prompt panel, every shared-protocol baseline trails SCA on all ten per-image metrics, while HRDiT and UltraImage lead on several preference metrics.** Baseline minus SCA (28 steps) at 4096^2 with 95% paired intervals; HPSv2, HPSv2.1, CLIP-IQA, and ARNIQA are $\times 100$, KID and KID_p $\times 10^3$. For FID, KID, their patch versions, and campaign time (seconds per image), lower is better, so negative favours the baseline. **Blue** / *orange*: baseline significantly better / worse than SCA. SigMa uses its dev-matched profile; UltraImage and HRDiT use their own recipes (Appendix G).

	Single pass, shared sampler				
Metric	Full attn.	DyPE	SigMa	SEGA	AReA
ImageReward	−3.338 [−3.532, −3.127]	−1.358 [−1.656, −1.065]	−1.895 [−2.224, −1.582]	−0.655 [−0.894, −0.436]	−0.291 [−0.434, −0.159]
Aesthetic	−2.13 [−2.30, −1.96]	−0.80 [−0.96, −0.63]	−0.93 [−1.11, −0.74]	−0.41 [−0.53, −0.30]	−0.11 [−0.18, −0.03]
CLIP score	−14.52 [−16.26, −12.74]	−4.55 [−5.77, −3.38]	−6.46 [−7.75, −5.22]	−2.05 [−2.78, −1.33]	−1.56 [−2.23, −0.87]
PickScore	−5.23 [−5.63, −4.82]	−2.71 [−3.03, −2.41]	−3.29 [−3.66, −2.92]	−1.40 [−1.70, −1.11]	−0.65 [−0.88, −0.43]
HPSv2	−9.30 [−10.02, −8.61]	−4.42 [−4.93, −3.95]	−5.22 [−5.83, −4.71]	−2.53 [−2.99, −2.10]	−1.02 [−1.33, −0.70]
HPSv2.1	−22.98 [−24.54, −21.24]	−9.74 [−11.01, −8.52]	−11.82 [−13.26, −10.46]	−5.16 [−6.19, −4.16]	−2.22 [−2.89, −1.54]
MUSIQ	−23.0 [−26.2, −19.5]	−9.3 [−13.9, −4.6]	−8.7 [−13.7, −3.5]	−9.6 [−12.8, −6.0]	−6.2 [−10.1, −1.9]
MUSIQ_c	−28.8 [−31.7, −25.9]	−11.6 [−15.2, −8.0]	−12.0 [−15.5, −8.7]	−11.2 [−15.0, −7.7]	−10.5 [−13.2, −7.6]
CLIP-IQA	−24.2 [−26.9, −21.4]	−16.1 [−19.9, −12.3]	−15.6 [−19.7, −11.5]	−7.2 [−10.7, −3.8]	−2.9 [−5.1, −0.7]
ARNIQA	−16.3 [−19.8, −12.8]	−11.2 [−14.0, −8.5]	−11.0 [−13.8, −8.3]	−6.3 [−8.7, −3.9]	−2.2 [−3.7, −1.0]
FID↓	+163.7 [+126.4, +178.2]	+76.0 [+52.8, +105.0]	+83.0 [+59.5, +105.2]	+36.9 [+17.7, +62.0]	+6.3 [−4.0, +17.4]
KID↓	+170.7 [+134.5, +207.8]	+46.9 [+30.8, +70.7]	+58.2 [+42.0, +81.6]	+6.4 [−1.6, +21.9]	+1.0 [−3.3, +5.5]
FID_p ↓	+124.4 [+85.5, +124.2]	+83.5 [+58.8, +100.2]	+87.4 [+62.1, +100.0]	+30.3 [+15.0, +45.2]	+0.4 [−9.8, +9.7]
KID_p ↓	+162.9 [+139.6, +184.8]	+54.4 [+35.4, +76.5]	+67.9 [+50.1, +88.8]	+12.2 [+3.5, +23.8]	−1.2 [−8.1, +4.0]
Time (s)↓	+120.0 [+116.0, +125.0]	+132.6 [+121.3, +145.0]	+113.9 [+113.3, +114.3]	+118.1 [+116.4, +120.0]	+169.4 [+156.2, +186.4]

	Multi-stage, own recipe	
Metric	UltraImage	HRDiT
ImageReward	+0.136 [−0.004, +0.270]	**+0.249** [+0.126, +0.378]
Aesthetic	+0.07 [−0.01, +0.15]	**+0.12** [+0.04, +0.21]
CLIP score	−0.64 [−1.26, −0.05]	−0.03 [−0.66, +0.60]
PickScore	**+0.35** [+0.17, +0.53]	**+0.46** [+0.28, +0.66]
HPSv2	**+0.36** [+0.05, +0.70]	+0.27 [−0.11, +0.66]
HPSv2.1	**+1.56** [+0.85, +2.28]	**+1.62** [+0.92, +2.36]
MUSIQ	−11.2 [−15.1, −7.0]	−5.1 [−8.5, −1.6]
MUSIQ_c	−19.6 [−22.4, −16.6]	−8.4 [−11.9, −5.2]
CLIP-IQA	+0.4 [−1.9, +2.7]	**+2.8** [+0.9, +4.8]
ARNIQA	**+2.1** [+0.3, +4.1]	+1.0 [−0.5, +2.6]
FID↓	+7.4 [−2.0, +17.7]	+10.5 [−0.6, +25.2]
KID↓	+2.0 [−2.6, +7.3]	+0.2 [−3.8, +7.5]
FID_p ↓	+14.2 [+5.5, +22.3]	+10.1 [+2.5, +20.0]
KID_p ↓	+12.0 [+7.1, +17.4]	+5.9 [+1.6, +10.9]
Time (s)↓	+250.4 [+235.6, +269.0]	+34.9 [+31.6, +39.4]

J SAMPLING-BUDGET AND GUIDANCE CHECKS

On the same 46 evaluation prompts, we compare the default 28-step SCA run with 56 steps at guidance 4.5 and with 28 steps at guidance 7 (Table 21). The 56-step run keeps the noise threshold at 0.89; it therefore changes the discretization of both the opening and mixed phases. It improves ImageReward by 0.097 $[0.039, 0.159]$, HPSv2.1 by 1.01, PickScore by 0.16, and MUSIQ$_c$ by 2.26, but increases corrected patch FID by 7.05 $[1.16, 13.03]$ with reference-image cluster resampling. Increasing guidance to 7 alone lowers HPSv2.1 and PickScore and does not resolve an ImageReward difference. Thus the default schedule does not maximize every metric, and extra sampling can change the preference-quality balance.

Against UltraImage on this subset, 56-step SCA has ImageReward difference -0.038 $[-0.159, 0.078]$, HPSv2.1 difference -0.56 $[-1.21, 0.09]$, and PickScore difference -0.19 $[-0.40, 0.01]$. These intervals span zero and do not establish equivalence. Its MUSIQ advantage is 9.94 $[5.69, 13.98]$ and its MUSIQ$_c$ advantage 21.83 $[18.67, 24.73]$. Campaign means are 71.6 s for default SCA, 140.1 s for 56-step SCA, and 322.0 s for UltraImage on these 46 prompts. This is a sampling-budget sensitivity check, not an equal-FLOP or equal-wall-clock experiment: UltraImage has two resolutions, different guidance, and per-prompt preparation. Against HRDiT on the same prompts (Table 2), the 56-step run stays lower on ImageReward (1.22 against 1.37), HPSv2.1 (30.99 against 31.60), and PickScore (23.12 against 23.42) while taking longer (140.1 against 106.5 s); these are point estimates without paired intervals. A 42-step run sized to HRDiT's campaign time and HRDiT's schedule with the SCA operator are pending. [TODO: 42-step and progressive SCA]

Table 21: **Doubling the steps raises seven metrics; guidance 7 lowers six.** On 46 prompts, the first two columns compare with default SCA; the last compares 56-step SCA with UltraImage. Per-image differences (first-named setting minus second) and 95% paired intervals, 2000 draws; **blue** / *orange*: the interval excludes zero in favour of / against the first-named setting.

Metric	56 vs. 28 steps	Guidance 7 vs. 4.5	56 steps vs. UltraImage
ImageReward	**+0.097** [+0.039, +0.159]	+0.019 [−0.070, +0.113]	−0.038 [−0.159, +0.078]
Aesthetic	+0.03 [−0.02, +0.09]	−0.10 [−0.18, −0.03]	−0.03 [−0.12, +0.05]
CLIP score	−0.32 [−0.92, +0.19]	−0.43 [−0.94, +0.01]	+0.32 [−0.43, +1.01]
PickScore	**+0.16** [+0.02, +0.31]	−0.27 [−0.41, −0.14]	−0.19 [−0.40, +0.01]
HPSv2 (×100)	**+0.34** [+0.13, +0.57]	−0.30 [−0.51, −0.10]	−0.02 [−0.34, +0.30]
HPSv2.1 (×100)	**+1.01** [+0.59, +1.48]	−0.74 [−1.08, −0.38]	−0.56 [−1.21, +0.09]
MUSIQ	−1.3 [−3.5, +1.0]	+1.3 [−1.2, +3.7]	**+9.9** [+5.7, +14.0]
MUSIQ$_c$	**+2.3** [+1.2, +3.4]	+1.0 [−0.2, +2.3]	**+21.8** [+18.7, +24.7]
CLIP-IQA (×100)	**+1.5** [+0.2, +2.7]	−1.5 [−2.9, −0.1]	+1.1 [−1.4, +3.4]
ARNIQA (×100)	**+1.3** [+0.3, +2.4]	−1.4 [−2.2, −0.7]	−0.8 [−2.9, +1.0]

K EXPLORATORY 8192^2 COMPARISON

Table 22 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol. SCA has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals excluding zero. Aesthetic score and the three remaining quality metrics have intervals spanning zero. In particular, MUSIQ, MUSIQ$_c$, and CLIP-IQA have point estimates favouring AReA. These mixed quality scores and the grid and stripe failures in Figure 20 limit the evidence for reliable 8K generation. With only 12 prompts, this comparison is exploratory; we do not report distributional metrics for this set.

All 8K scores were computed on CPU. HPS uses float32 here, whereas the main GPU evaluation uses mixed precision, so absolute scores across resolutions should not be interpreted as a controlled scaling comparison. Both methods within this table use the same evaluator settings. For native-resolution MUSIQ, attention queries are processed in chunks of 256 while retaining every key and native image patch. This bounds intermediate memory without changing the attention context; pre-trained equivalence checks at three smaller image sizes reproduced the unchunked scores exactly.

Table 22: **Exploratory 8192^2 comparison: SCA is significantly better than AReA on six of ten metrics.** FLUX.1-dev, 12 matched prompts, seed 0. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2000 draws, bootstrap seed 0); **blue**: significantly better than AReA.

Metric	AReA	SCA	SCA − AReA [95% CI]
ImageReward	−0.376	0.675	**+1.051** [+0.386, +1.754]
Aesthetic	5.64	5.72	+0.08 [−0.26, +0.44]
CLIP score	22.38	25.52	**+3.14** [+0.63, +6.11]
PickScore	20.35	21.27	**+0.92** [+0.24, +1.75]
HPSv2 (×100)	24.60	26.13	**+1.53** [+0.56, +2.66]
HPSv2.1 (×100)	16.59	20.51	**+3.92** [+1.31, +6.85]
MUSIQ	28.6	25.7	−2.9 [−6.8, +0.8]
MUSIQ_c	44.0	40.1	−3.8 [−10.4, +1.6]
CLIP-IQA (×100)	57.1	52.0	−5.1 [−10.5, +0.6]
ARNIQA (×100)	37.3	54.6	**+17.3** [+11.9, +22.6]

L CALIBRATION TABLE AT NON-SQUARE SIZES

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, the scale-2 table changes the paired metrics negligibly (Table 14): HPSv2 +0.09, HPSv2.1 +0.17 (both ×100), and PickScore +0.05 are significant but small, and ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. These small changes do not resolve the mixed SCA–AReA results at this size, and we keep the default rule.

M THE LOW-BAND DUPLICATION GUARD

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\text{HP}_p = \text{I} - \text{LP}_p$. The guard replaces the low band of the mix M by that of the global branch:

$$O \;=\; \text{LP}_p\big(O^{\text{glob}}\big) + \text{HP}_p\big(M\big) \;=\; O^{\text{glob}} + \pi_h(\sigma)\,\text{HP}_p\big(O^{\text{loc}} - O^{\text{glob}}\big). \tag{6}$$

The local branch then contributes only its difference from the global branch above the cutoff. The cutoff should lie below $1/(2s)$, where the global branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size. ResDiT also combines global low frequencies and local high frequencies, using patch-wise FFT masks (Ma et al., 2025). Our optional guard uses pooled global attention and preserves the fitted per-head weight on the local-minus-global high-band residual. Its global-low/local-high construction is therefore not itself novel.

History: what it fixes and what it costs. On the development prompts, the band alone (no opening phase) fixed the boats but left extra statues and faces on the Moai and woman-and-bear prompts; the opening phase alone got the main subject right, but small extra figures appeared late on some prompts; both together gave one subject on the boats, Moai, and woman-and-bear prompts. This was our first full configuration. On 46 prompts, however, removing the guard improves HPSv2 (+0.54), HPSv2.1 (+1.17), PickScore (+0.25), CLIP-IQA (+1.7), and MUSIQ (+2.9), all significant, leaves the other metrics within noise, and saves 17 s per image on an idle GPU. The guard also hid what the per-head mix contributes. With it (Table 23), local only matched the per-head mix within 2% on preference metrics, because the guard replaces the low band of every head's output with that of the global branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 4.2). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold

Table 23: **Ablations around the band-guard configuration at 4096^2**, 46 prompts. The reference row gives absolute values for SCA + band guard (band 16); every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 3; corrected patch FID is shown without intervals.

Variant	Mix err.	Image Reward	Aesth.	HPSv2	HPSv2.1	Pick Score	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_c	$\text{FID}_p \downarrow$
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	166.7
no band (**= SCA final**)	0.200	−0.04	+0.01	**+0.54**	**+1.17**	**+0.25**	+0.40	**+1.7**	+0.9	**+2.9**	+0.4	+1.3
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+8.9
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.6
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.3
local only ($\pi = 1$)	0.261	−0.01	+0.03	**+0.16**	**+0.55**	**+0.12**	+0.16	+0.5	+0.6	+1.0	**+0.9**	+1.6
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	−0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	**+0.34**	**+0.09**	+0.13	+0.6	+0.6	+0.8	**+0.8**	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.7

0.5, about 6% at 0.3) stayed within 2% of the mix. The guard is an option for prompts where late duplication is visible, not part of the default method.

N Compute and reproducibility

Compute. Generation and calibration used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120 GB model. Some metric evaluations ran on CPUs. The surviving logs do not establish the complete calibration runtime. Generation at 4096^2 takes about 71 s per image for SCA, 262 s for AReA, and 197 s for full attention on shared nodes (campaign means); the main comparisons, ablations, and non-square sizes comprise more than 2000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours through 2026-10-17 06:59 UTC, including model loading, calibration, evaluation, and failed attempts. The bound is the union of 581 closed task intervals per physical node, with timestamp precision treated conservatively; it measures occupied worker time, not GPU kernel activity. Missing historical logs, unmatched starts, and ongoing work prevent a complete campaign total. Overlapping scheduler allocation time and image-sidecar durations are not added to this bound.

Reproducibility. The code (attention processor, calibration, sampling, evaluation wrappers, AReA wrapper, and tests), the calibration table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table will be released. Main-text metric values and paired intervals are derived from those CSV files. Each SCA generation task ran from a frozen copy of the code named in the task. Per-image sidecar files record the sampling settings and, where available, a code fingerprint; the inherited Krea baseline records do not establish one source revision for every image.

O Broader impact

SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

P QUALITATIVE GALLERIES

Figure 11 shows curated SCA outputs. The other galleries, Figures 12 to 20, are uncurated unless stated: prompts are drawn at random from the evaluation prompts, excluding the development prompts and those shown in the main text.

Figure 11: **SCA keeps layout and fine detail across three aspect ratios without training.** Curated SCA outputs at 4096^2, 2048×4096 and 2048×6144 on evaluation prompts, picked by eye for variety (FLUX.1-dev, seed 0, 28 steps, guidance 4.5). Insets are native-pixel crops of the outlined windows.

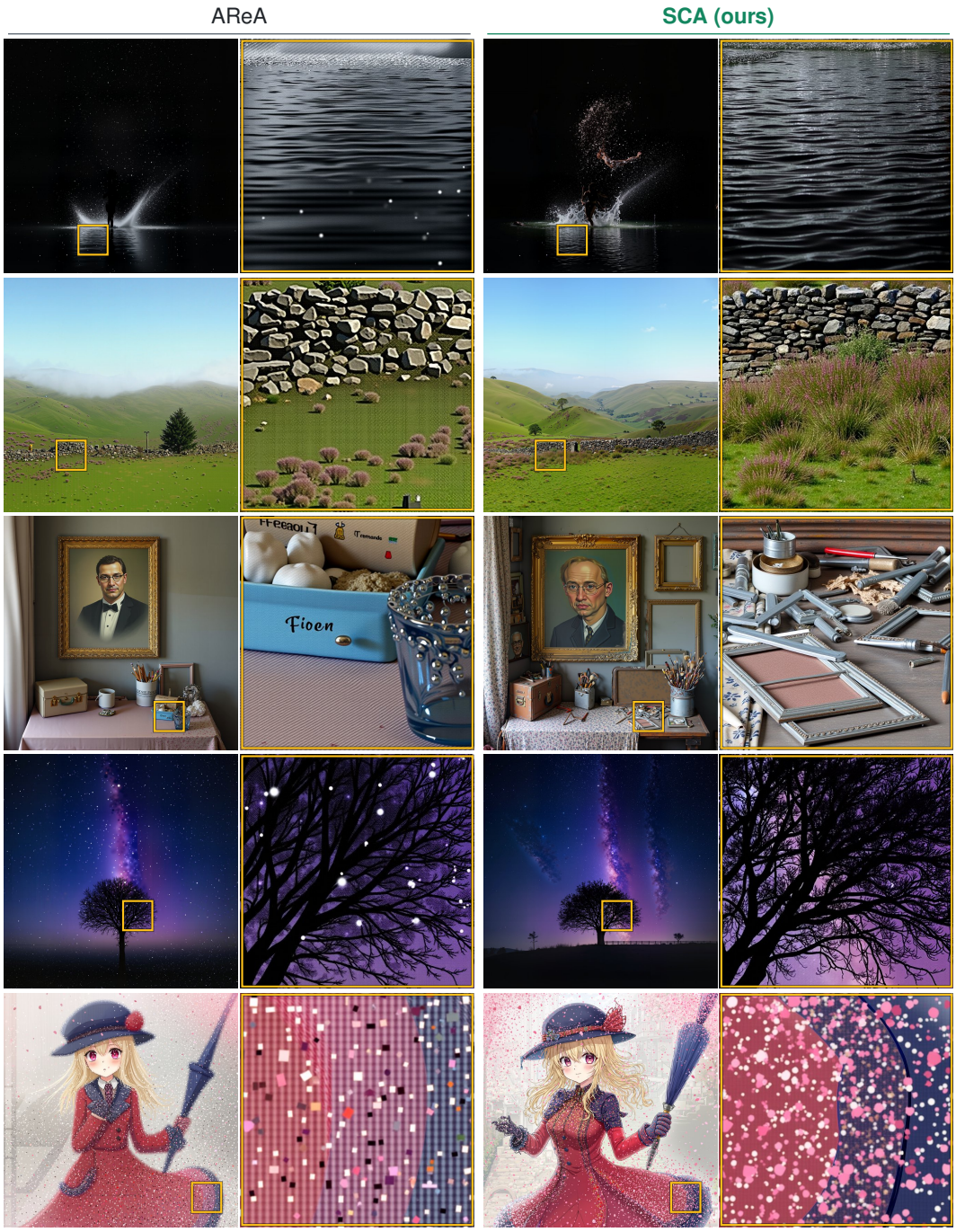

Figure 12: **Uncurated samples at 4096×4096: AReA and SCA (ours).** The first five of 32 evaluation prompts drawn at random (selection seed 0), excluding development and main-text prompts. FLUX.1-dev, seed 0, 28 steps, guidance 4.5. Each image is followed by a native-pixel crop of its outlined 512×512 window of highest edge energy averaged over the methods.

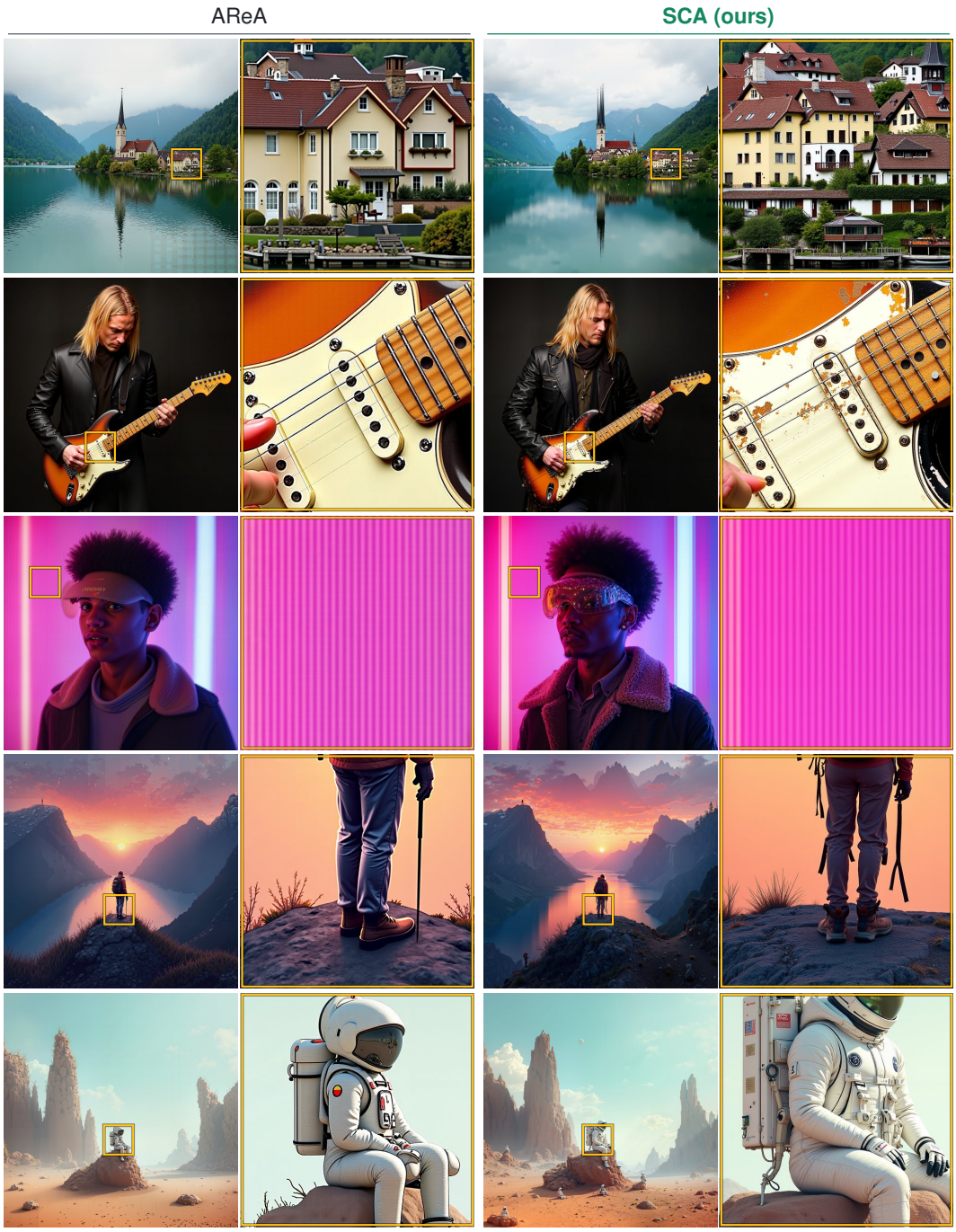

Figure 13: **Uncurated samples at 4096×4096: AReA and SCA (ours), continued.** Prompts six to ten of the same draw; same protocol as Figure 12.

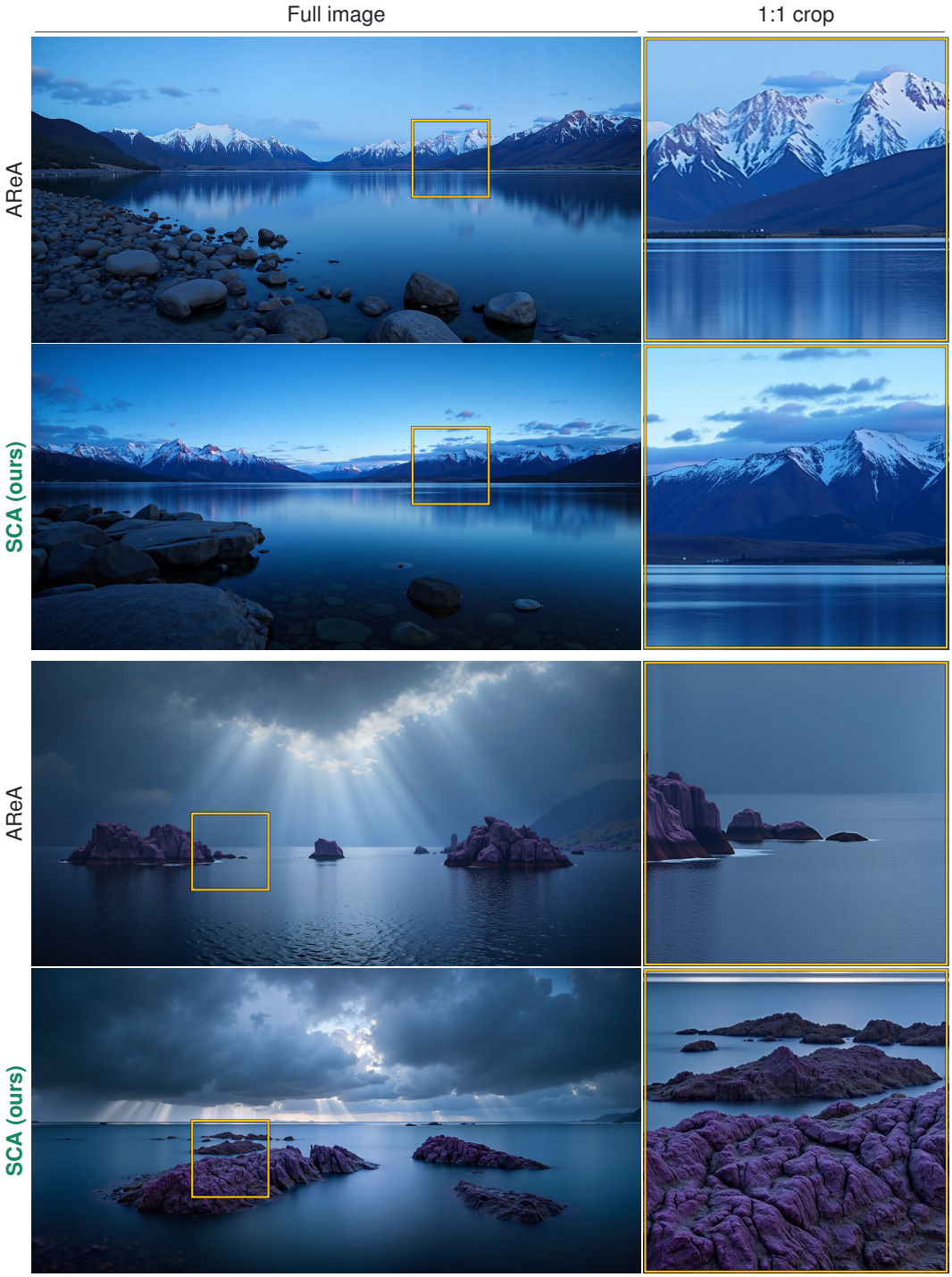

Figure 14: **Uncurated samples at 2048×4096 (height×width): AReA and SCA (ours).** The first two of six evaluation prompts drawn at random, disjoint from the other galleries, excluding development and main-text prompts. FLUX.1-dev, seed 0, 28 steps, guidance 4.5. Crops are native-pixel views of the outlined 512×512 window of highest edge energy averaged over the methods.

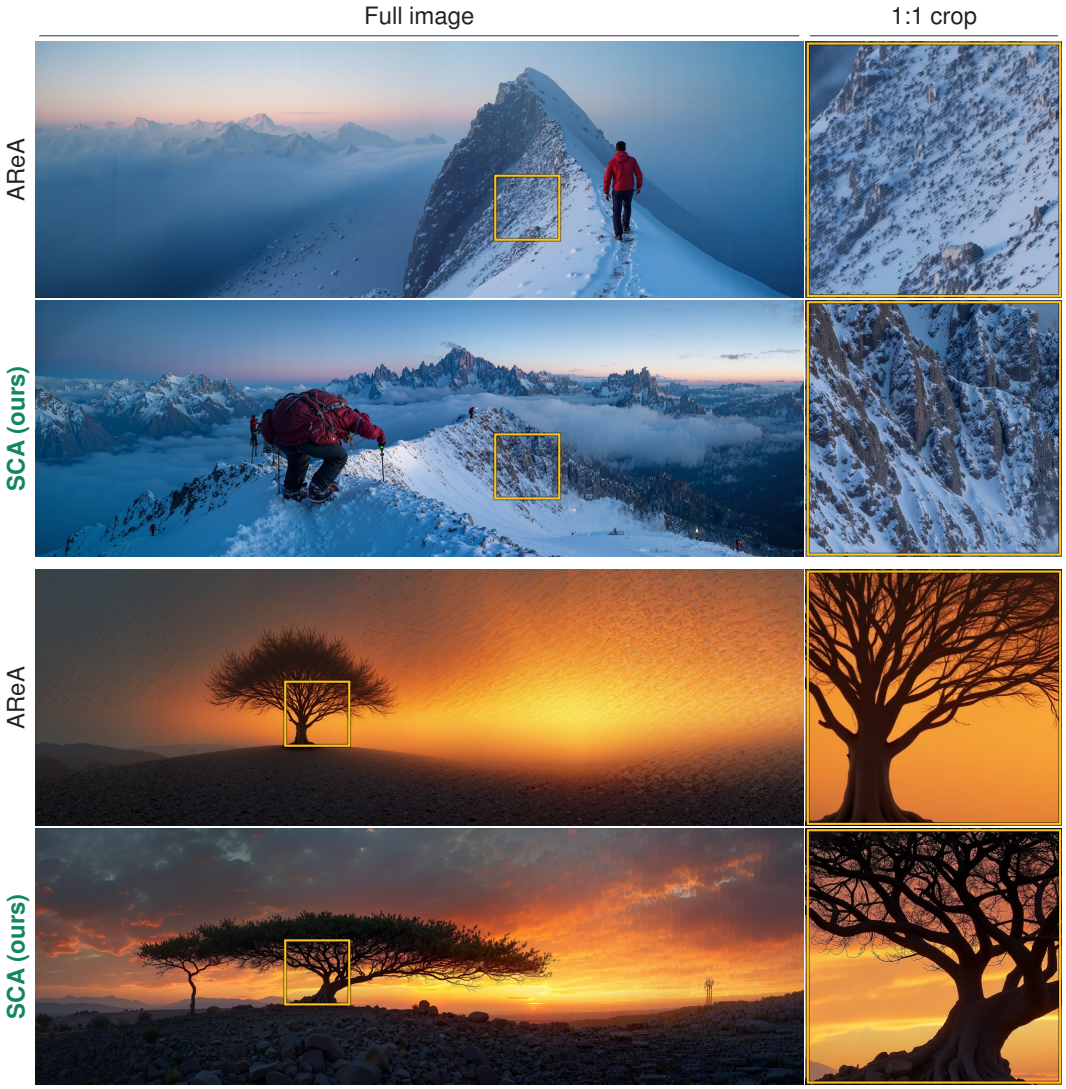

Figure 15: **Uncurated samples at 2048×6144 (height×width): AReA and SCA (ours).** The first two of six evaluation prompts drawn at random, disjoint from the other galleries, excluding development and main-text prompts. FLUX.1-dev, seed 0, 28 steps, guidance 4.5. Crops are native-pixel views of the outlined 512×512 window of highest edge energy averaged over the methods.

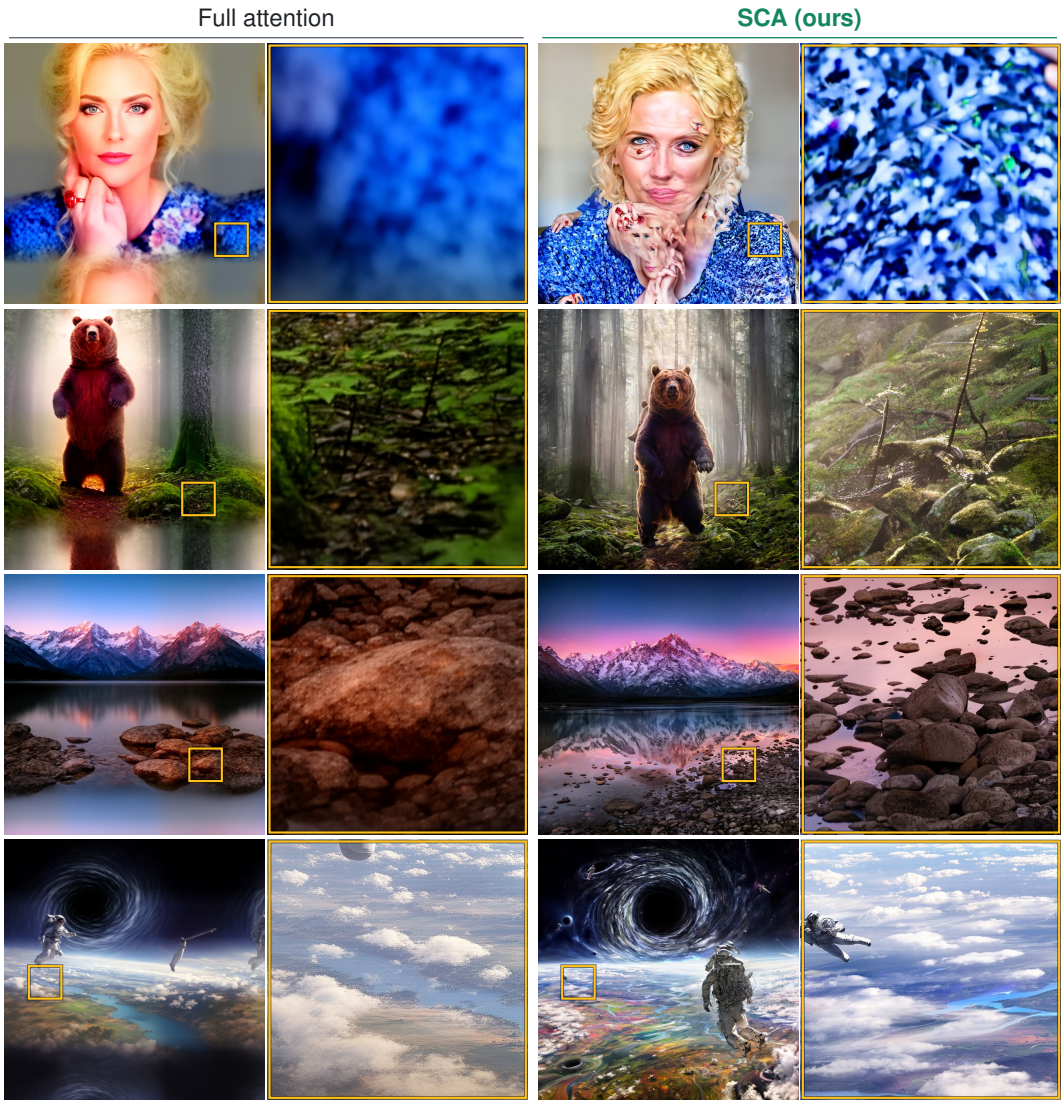

Figure 16: **FLUX.2-dev at 4096^2: SCA can still distort face geometry and fine texture (first row).** Four of the 50 paired evaluation prompts, sampled at random (selection seed 17), in draw order. Seed 0, 28 steps, guidance 4.0, FLUX.2 calibration table for SCA; each image is followed by a native-pixel crop of its outlined window.

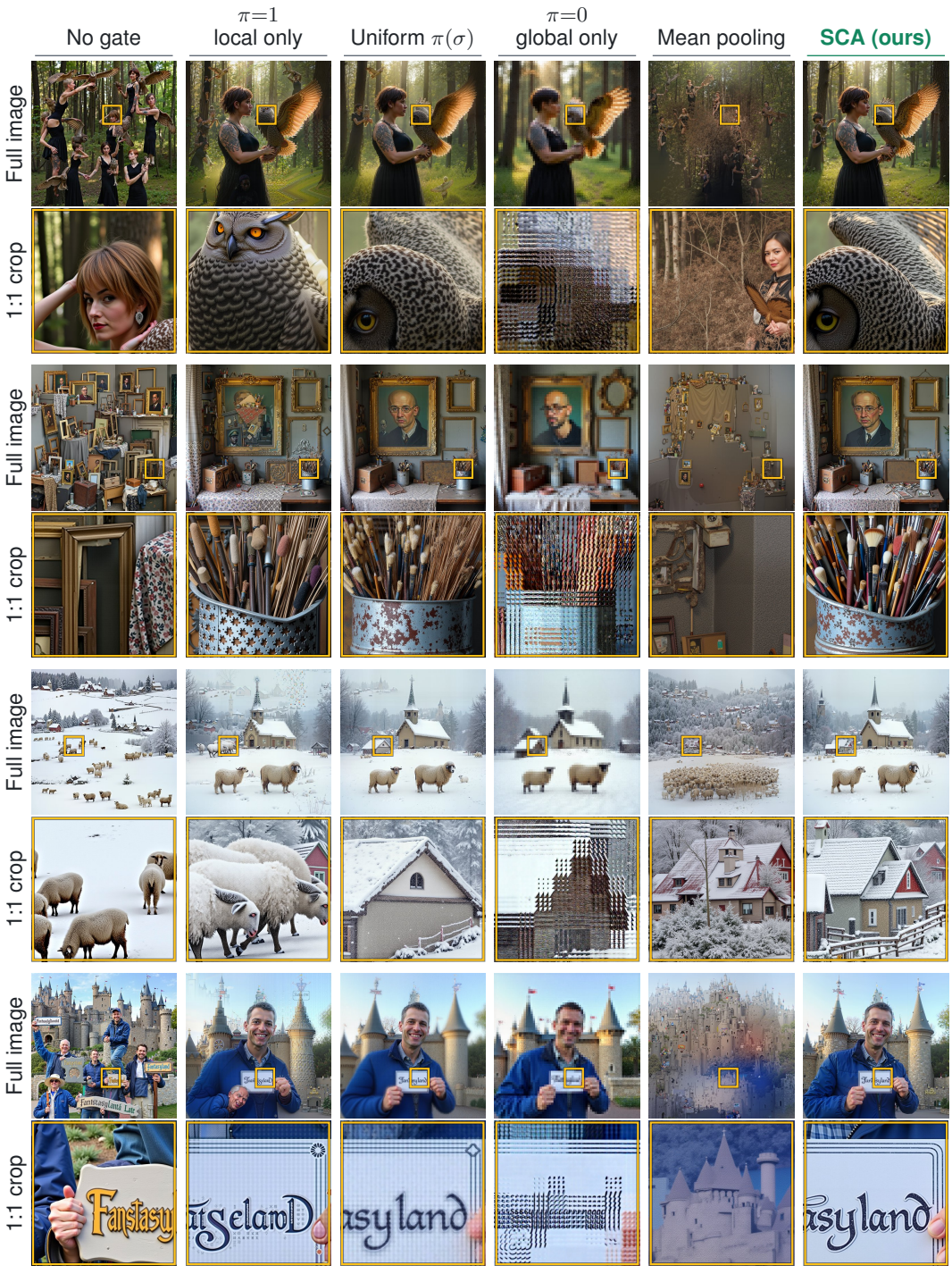

Figure 17: **Without the gate the subject repeats; $\pi = 0$ leaves the pooled branch's coarse grid, and mean pooling loses the layout.** Four ablation prompts at 4096×4096, seed 0, 28 steps, guidance 4.5; variants as in Table 3; crops show native pixels of the outlined 512×512 windows. $\pi = 1$ and uniform $\pi(\sigma)$ look close to SCA here; their deficits show in the metrics.

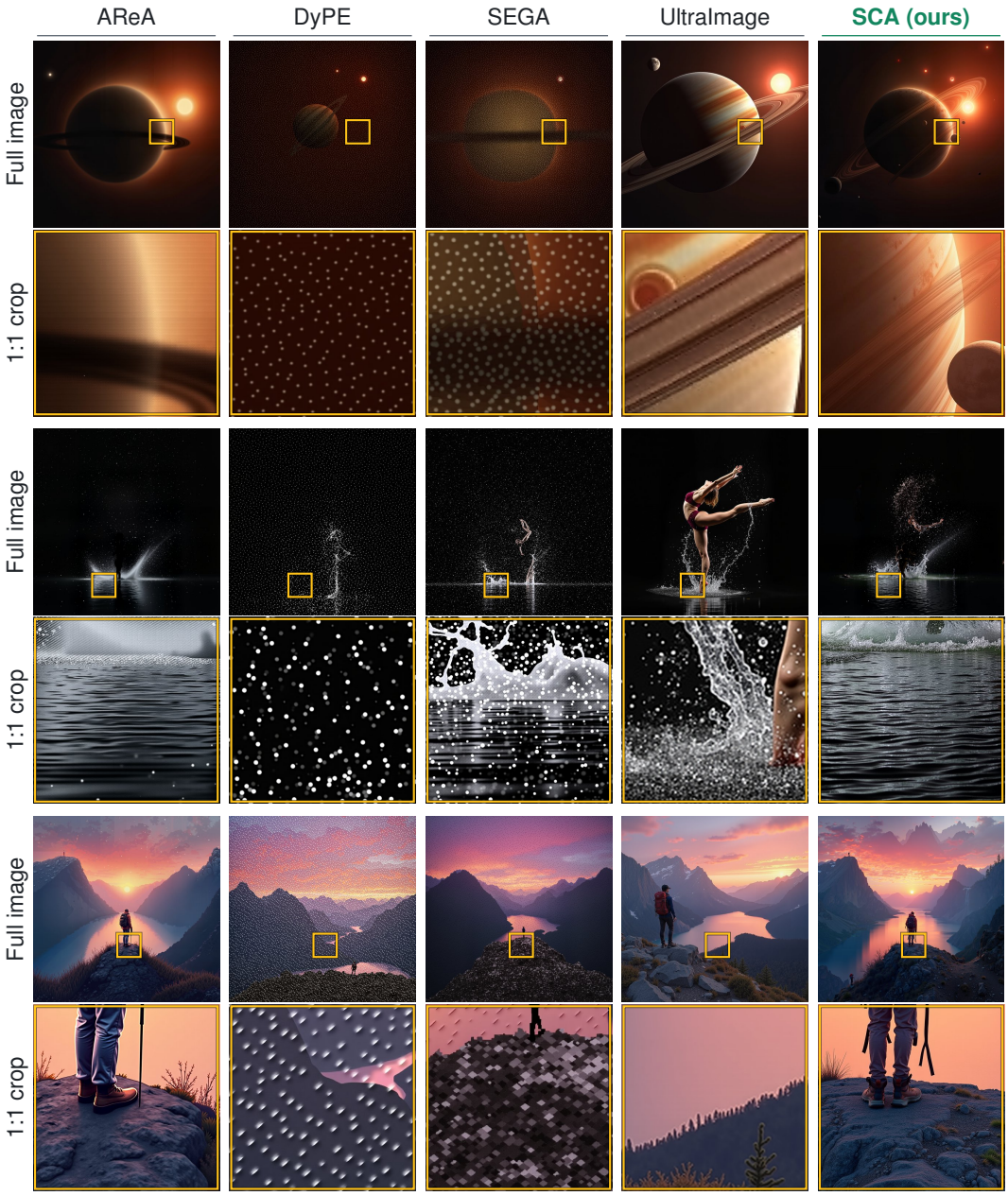

Figure 18: **Additional baselines at 4096^2.** Three of the 191 paired evaluation prompts, sampled at random (selection seed 17); each image sits above a native-pixel crop of its outlined window. FLUX.1-dev, seed 0. AReA, DyPE, SEGA and SCA use 28 steps and guidance 4.5; UltraImage uses its two-stage defaults (30+30 steps, guidance 3.5 then 7), so this is not an equal-cost comparison.

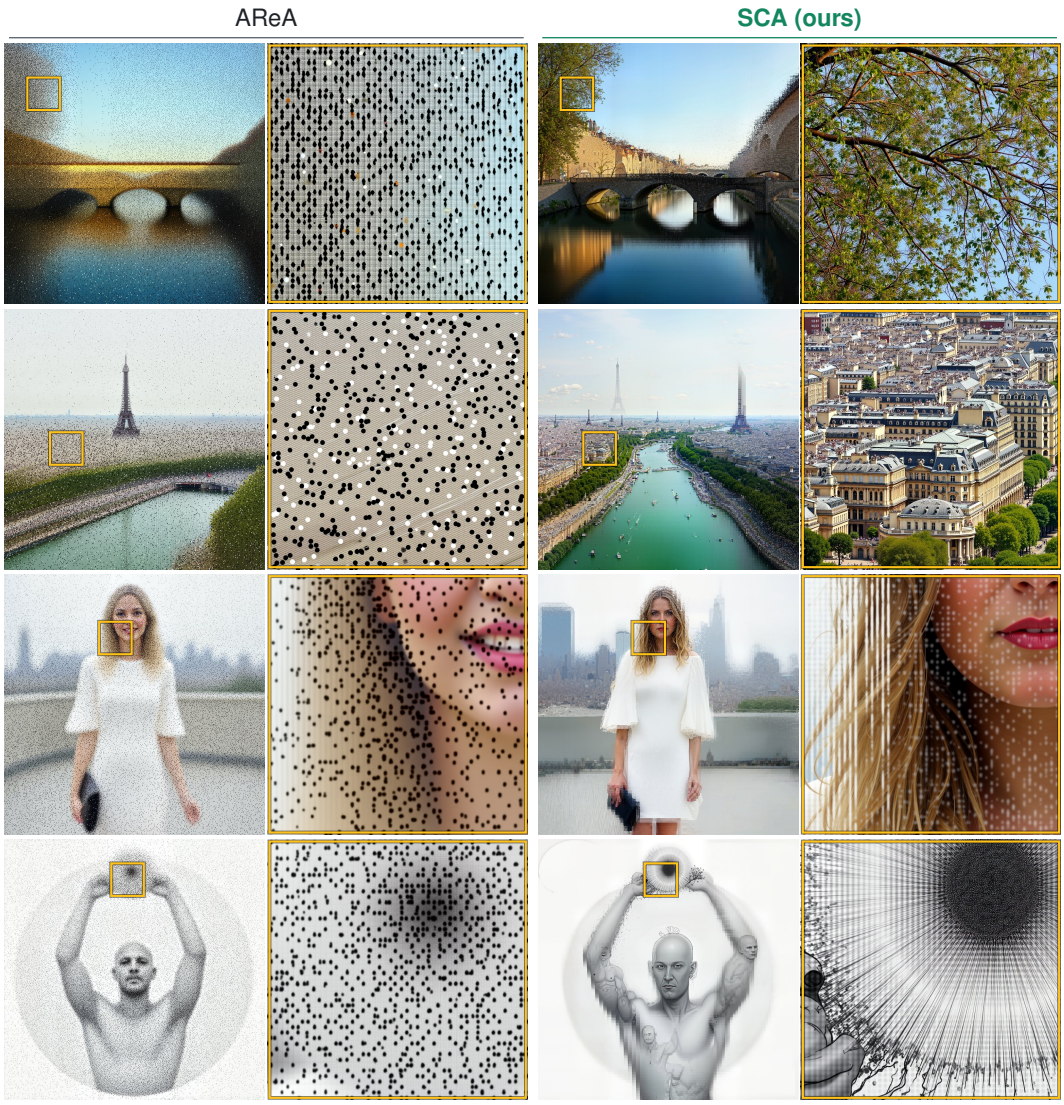

Figure 19: **Partial qualitative results at 6144^2.** Four prompts sampled at random (selection seed 17) from the 14 non-development prompts with completed AReA and SCA outputs at the figure snapshot, a pool that may reflect completion order; this is not the final evaluation set. FLUX.1-dev, seed 0, 28 steps, guidance 4.5; each image is followed by a native-pixel crop.

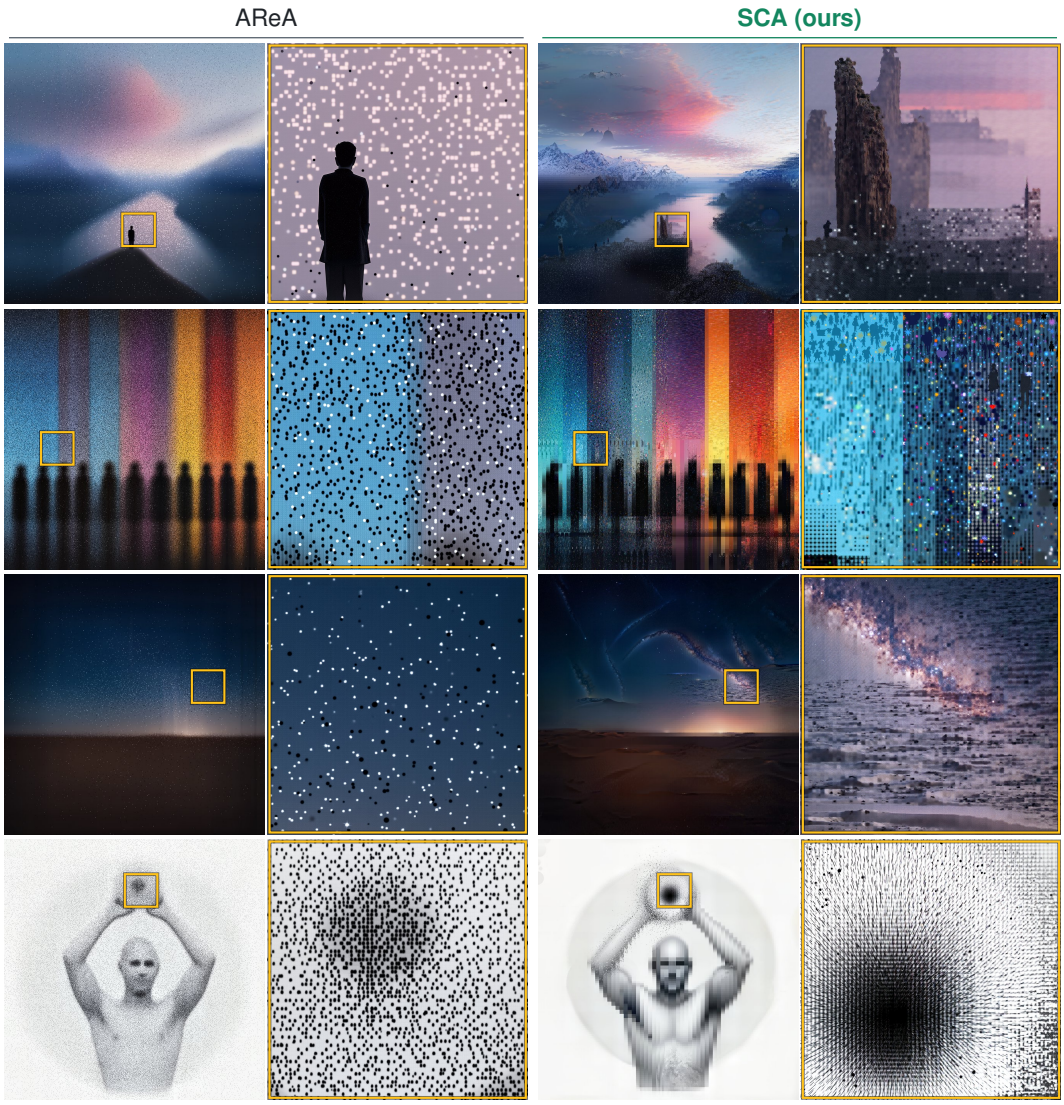

Figure 20: **Exploratory 8192^2 generation: grid and stripe artifacts remain at this scale.** Four of the 12 paired evaluation prompts, sampled at random (selection seed 17). FLUX.1-dev, seed 0, 28 steps, guidance 4.5; each image is followed by a native-pixel crop of its outlined window. This small evaluation does not establish general 8K quality.

Q FAILURE CASES

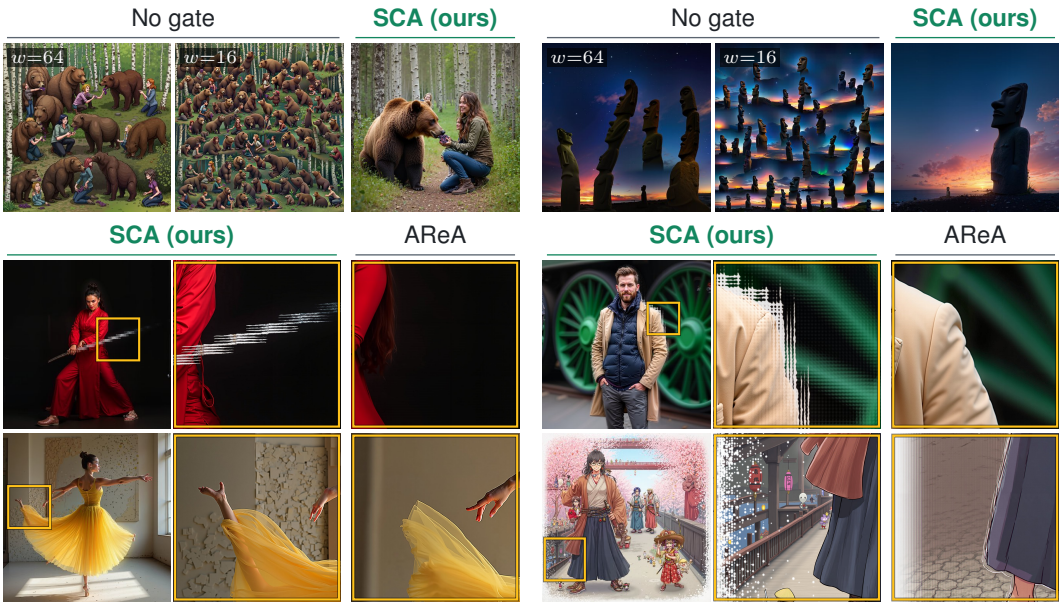

Figure 21: **Without the gate, windows duplicate the subject; SCA still makes local errors.** Top (development prompts): no gate with 64- and 16-token windows (w), then SCA. Below: a staircase sword, a grid beside a coat, a foot for an arm, border speckle, each next to AReA's same window; AReA composes differently and avoids them. 4096×4096, seed 0, 28 steps, guidance 4.5.

Figure 21 shows the duplication that windows cause without the gate, and local errors that SCA still makes.

Late duplication. With the final method, small extra figures can appear on some prompts in the last 20 steps, when the local branch dominates most heads. The band guard suppresses some of them but costs preference scores.

Articulated subjects. The underwater-surfer development prompt keeps doubled limbs in every variant; AReA's image shows a large translucent ghost.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis, and the per-head mix gains little over AReA's shrinking windows; aesthetic score and corrected patch FID are significantly worse.

Larger scales and another backbone. The gallery sampled from the completed 12-prompt 8192^2 run still shows grid and stripe artifacts in both methods (Figure 20); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distorted facial geometry and fine texture in the randomly selected portrait example (Figure 16).

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharp repeated subjects (v0 scores higher on CLIP-IQA and ARNIQA), and FID against 195 references at 299^2 resolution sees little of the 4K detail; we therefore report every metric and the qualitative galleries.